

CARBON FIBRE TECHNOLOGY AND ADVANCED COMPOSITES IN THE INDIAN SPORTS SECTOR

*Study on Global Performance Benchmarks, Multi-Industry Cost
Modelling, and Manufacturing Policy Frameworks*

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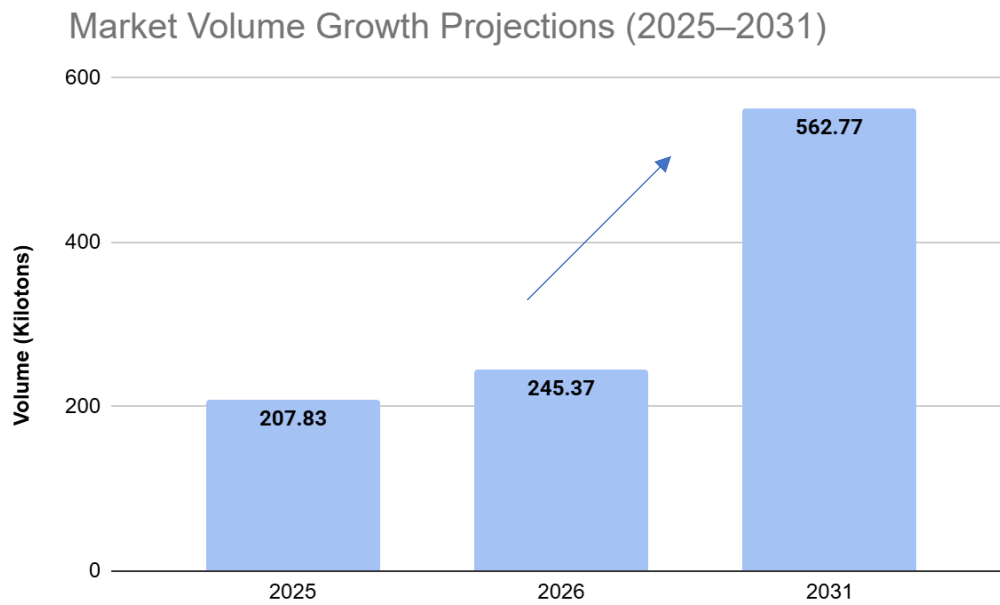
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Chapter 1: Executive Summary

The consistent pursuit of superior athletic performance has long been a primary driver of innovation in sports equipment design and materials engineering. Over the past two decades, composite materials have established themselves as the material of choice for high-performance sports applications, delivering combinations of strength, stiffness, damping characteristics, and weight reduction that conventional metallic materials are fundamentally unable to match. The global sports composites market, valued at approximately \$3.82 billion in 2023, is projected to reach \$5.67 billion by 2030, reflecting the accelerating adoption of advanced materials across diverse sporting disciplines. In sports equipment applications, the most common configuration involves continuous fibre reinforcements, typically carbon, glass, or aramid, embedded in a polymer matrix such as epoxy, polyester, or thermoplastic resins. This architecture enables directional property optimisation, where fibres can be strategically oriented to resist dominant loading patterns while minimising weight in non-critical directions. Carbon fibre has emerged as one of the most consequential advanced materials of the twenty-first century. Its defining properties, a tensile strength exceeding that of steel at roughly one-fifth the weight, high stiffness, corrosion resistance, and thermal stability, have made it indispensable across industries that demand performance under extreme conditions.

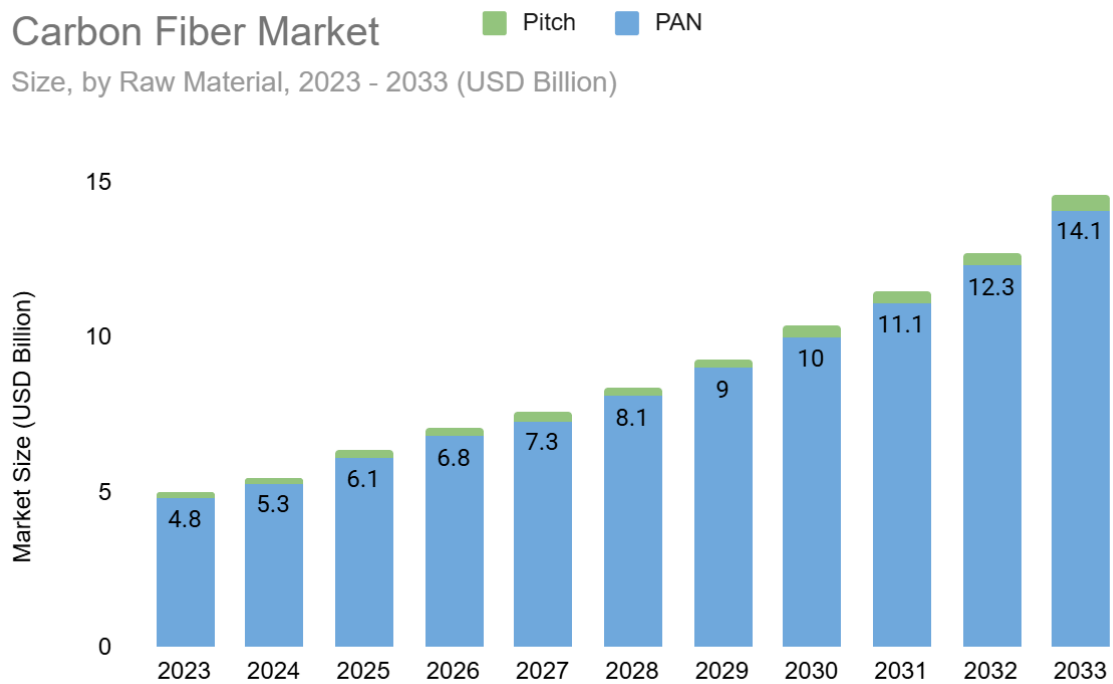
Globally, the market is on a steep upward trajectory. In volume terms, the carbon fibre market is projected to grow from approximately 245 kilotons in 2026 to nearly 563 kilotons by 2031, representing a CAGR of 18.06% (Figure 1) driven by surging demand from clean energy, electric vehicles, and aerospace programmes. In value terms, the market is expected to expand from USD 6.41 billion to USD 9.07 billion over the same period, reflecting a 7.2% CAGR as pricing per kilogram moderates with scale. This divergence between volume and value growth is itself significant: it signals a material transition from premium niche to industrial commodity, with manufacturing accessibility becoming the central competitive frontier.

Figure 1 Global Carbon Market Growth



Source: Mordor Intelligence

Figure 2 Carbon fibre market in USD Billion



Source: [grandviewresearch.com/industry-analysis/carbon-fibre-market-analysis](https://www.grandviewresearch.com/industry-analysis/carbon-fibre-market-analysis)

Within this global shift, India occupies a particularly consequential position. As the world's fifth-largest economy with a rapidly expanding manufacturing base, a stated national commitment to aerospace self-reliance under initiatives such as Make in India and Atmanirbhar Bharat, and a growing domestic market for sporting goods, automotive components, and renewable energy infrastructure, India presents both a compelling demand story and an underexplored manufacturing opportunity. Yet despite these tailwinds, India remains almost entirely dependent on imports for carbon fibre and its precursor materials, primarily polyacrylonitrile (PAN), with no significant domestic production capacity at scale. This gap between demand potential and supply capability forms the central problem this research seeks to examine.

1.1 The Three Generations of Sports Composites

The transformation of sports equipment from basic natural materials to highly engineered performance gear can be mapped across three distinct generational phases. This evolution highlights a continuous quest for optimal strength, minimal weight, and highly tuned biomechanical efficiency.

The First Generation (Pre-1970s)

For centuries, the sporting world relied almost exclusively on naturally occurring materials, with wood being the undisputed standard for equipment such as cricket bats, hockey sticks, tennis rackets, and championship rowing shells. While wood offered a natural flex and familiar feel, it was plagued by inherent performance inconsistencies, susceptibility to moisture warping, and a high likelihood of breaking under severe impact. To overcome these limitations, the sports industry initiated its first major material transition by adopting metals like steel and aluminium. Aluminium, in particular, offered significantly improved durability and was used to manufacture the first metal hockey shafts in the 1990s. This metal era was punctuated by controversial milestones, such as in 1979 when Australian cricketer Dennis Lillee famously used a solid aluminium cricket bat, 'The Combat', during a test match event that forced regulatory changes but clearly demonstrated the industry's desire to move past the limitations of wood.

The Second Generation (1970s–1990s)

The second generation marked the experimental introduction of carbon fibre into sports, following its early development for aerospace and defence applications. After Dr Akio Shindo developed polyacrylonitrile (PAN)-based carbon fibre in the early 1960s, manufacturers slowly began applying the technology to consumer goods. The cycling industry saw some of the earliest adaptations; the very first carbon fibre bicycle was unveiled in 1971, and by 1976, the U.S. Olympic team competed on the 'Graftek G-1', a hybrid frame featuring carbon fibre bonded to aluminium lugs. However, these early tube-to-tube bonding methods were often inconsistent and prone to joint failure.

A major engineering leap occurred in the mid-1980s. In 1983, the Specialized Epic became the first mass-produced carbon bike, and in 1986, the Kestrel 4000 introduced the first true 'monocoque' (single-piece moulded) carbon bicycle frame, allowing for aerodynamic shapes that were impossible to achieve with metal. During this same era, adaptive sports experienced a revolution when inventor Van Phillips studied the biomechanics of cheetahs and diving boards to design the first 'Flex-Foot' running blade, eventually leading to the launch of the iconic carbon fibre 'Cheetah' blade by Össur in 1996.

The Third Generation (2000s–Present)

The turn of the 21st century ushered in the third generation of sports composites, an era defined by the shift from early, experimental carbon fibres to ultra-high-modulus grades (such as T800 and T1100G) and advanced resin systems. This period is seeing carbon fibre transition from an 'exotic' material into the dominant gold standard for elite competition. In team sports, the early 2000s marked the revolution of the fully composite, one-piece hockey stick, completely replacing wood and aluminium hybrids.

The third generation has also been heavily characterised by the concept of 'technological doping' due to unprecedented performance enhancements. The Sydney 2000 and London 2012 Paralympic Games showcased highly individualised and ultra-lightweight composite racing wheelchairs that drastically improved mechanical efficiency and energy return. In 2017, Nike introduced the Vaporfly running shoe, embedding a full-length, rigid carbon fibre plate, acting as an energy-returning lever, this advancement improving running economy by up to 1.5% and permanently altered the landscape of competitive marathons. Today, third-generation manufacturing utilises robotic layups, targeted multi-directional weaves, and digitally modelled geometries, with an increasing focus on integrating recyclable thermoplastic resins to address the environmental footprint of modern sports gear.

1.2 The Global Carbon Fibre Sports Market -Scale, Growth, and Key Players

The commercial scale of carbon fibre in sport reflects the breadth of its adoption. The global carbon fibre sports market is projected to grow at a CAGR of 6.9% from 2025 to 2035, driven by rising demand for lightweight, high-performance materials across cycling, golf, racket sports, and high-performance athletics. Technological advances in manufacturing are simultaneously improving product performance and reducing production costs, accelerating adoption across a wider range of sports and price segments.

Regionally, North America and Europe currently dominate the market, supported by high consumer awareness and the concentration of key manufacturers. However, the Asia-Pacific region is emerging as the most significant growth frontier, driven by rising sports participation, infrastructure investment, and a growing middle class with increased spending on premium sporting goods trends that carry direct implications for India (Table 1).

Table 1 Regional Market Overview and Growth Drivers

Region	Market size (\$ bn)	Growth drivers
Asia-Pacific	1.1	Robust manufacturing, industrialisation, and cost competitiveness
Europe	0.6	Strong aerospace & wind energy industries, stringent regulations
North America	0.5	Aerospace, defence, and premium automotive segments
Latin America	0.07	Rising infrastructure investment, niche sporting goods demand
Middle East & Africa	0.03	Emerging construction and energy sectors

Source: Thryve Research

The market is led by a small number of large-scale global manufacturers. Toray Industries, the world's largest carbon fibre producer, announced in June 2024 the acquisition of Park Aerospace Corp's carbon fibre prepreg assets, strengthening its position in sports-grade composites. Hexcel Corporation entered a co-development partnership with SGL Carbon in April 2024 to produce next-generation carbon fibre fabrics for premium sports equipment. Teijin Limited launched a new carbon fibre prepreg specifically designed for high-performance cycling frames in February 2024. Other significant players include Mitsubishi Chemical Holdings, Solvay SA, ZOLTEK, Cytec Industries, and Mitsubishi Rayon a concentrated supply chain that currently presents both a challenge and an opportunity for emerging manufacturing nations such as India.

Table 2: Global Carbon Fibre Industry: Leading Producers Analysis

Company Name & HQ	Est. Year	Core Products	Key Strengths	Key Challenges
Toray Industries, Inc. Tokyo, Japan (Chuo-ku)	1926	Carbon fibre fabrics, resin systems, composites	Commands ~30% global market share; major supplier for Boeing/Airbus, EVs, and medical gear; operates 200+ facilities.	Elevated manufacturing costs due to strict environmental rules, and facing intense competition in Asia.
Teijin Limited	1918	Carbon fibre prepregs, aramid	Operates 169 global subsidiaries (115	Exposed to global currency exchange

Tokyo, Japan (Chiyoda-ku)		fibres, composite resins	overseas); holds gold-level eco-certification; advanced R&D.	risks; faces fierce competition from Toray and Mitsubishi.
Hexcel Corporation Stamford, CT, USA	1948	HexTow® carbon fibre, HexWeb® honeycomb cores, prepregs	Global leader in aerospace honeycomb manufacturing; holds 1,350+ patents; primary supplier for Boeing 787 and F-35.	Vulnerable to long aerospace market cycles; price pressures from Asian manufacturers.
Mitsubishi Chemical Holdings Tokyo, Japan (Chiyoda-ku)	2005	PYROFIL® carbon fibre, thermoplastic prepregs	Broad product portfolio across chemicals and healthcare; highly secure raw material access paths.	Dropping profit margins in lower-tier chemical lines; highly vulnerable to raw material price shifts.
SGL Carbon SE Wiesbaden, Germany	1878	Carbon fibre thermoplastics, glass-carbon composites	Controls full value chain (graphite to carbon fibre); supplies Airbus A359; runs 29 global production bases.	Suffered a 2024 profit dip due to high energy and tech costs; needs to scale up composite market share.
Toho Tenax Co., Ltd. Tokyo, Japan (Bunkyo-ku)	1934	Tenax carbon fibre, Bes Fight series	Maintains a 13,500-ton annual capacity; fully owned by Teijin, allowing for shared R&D and global reach.	Over-reliant on carbon fibre lines with minimal diversification, revenues swing with market shifts.
DowAksa Advanced Composites Yalova, Turkey	2012	High-performance carbon fibre, polyurethane-carbon fibre tech	Backed by a strong joint venture legacy (Dow + Aksa), highly competitive in defence and transit sectors.	Hit by strategic uncertainty following Dow's 2025 exit; slow corporate decision-making inherited from JV setup.

Hyosung Corporation Seoul, South Korea	1917	Carbon fibre composites, GV300S motorcycles	Dominates the Korean motorcycle industry; possesses mature composite technology; operates in 29 countries.	Premium production costs limit access to low-end markets; slow to adjust to fast global industry shifts.
Zhongfu Shenyang Carbon Fibre Co. Jiangsu, China	2006	High-strength (SYT) & high-modulus (SYM) carbon fibres, prepregs	Only Chinese firm utilizing dry-spray wet-spinning tech; 2023 capacity reached 28,500 tons (top 10 globally); AS9100D aerospace certified.	Needs to increase its share of high-end product lines; international brand recognition lags behind Japan/US peers.
Hebei Carbon Valley Carbon Fibre Co. Hebei, China	2020	Pre-oxidized & woven fabrics, carbon/carbon preforms, prepregs	Central player in the Beijing-Tianjin-Hebei industry base; 2024 phase-2 expansion targets 200,000 pieces annually; defense partner (C929 aircraft).	Very short company history with low global market penetration; needs to overcome technical limits in high-performance fibres.

Source: Industry Data Source: Top 10 Global Carbon Fibre Manufacturers 2025 Industry Analysis. Carbon Fibre Material. <https://www.carbonfibrematerial.com/top-10-global-carbon-fibre-manufacturers-2025-industry-analysis/>

Figure 3: Top Carbon fibre manufacturers



1.3 Scope and Objectives of This Report

This report examines the rise of carbon fibre technology in the context of Indian sport, with attention to three interconnected dimensions: athletic performance, market accessibility, and domestic manufacturing potential. It draws on global research in advanced composite materials, India-specific industry and policy data, and the broader literature on sports technology and economic development.

The report is structured to serve multiple audiences. For policymakers, it outlines the structural barriers and enablers that will determine whether India becomes a meaningful participant in the global carbon fibre sports equipment value chain. For manufacturers and investors, it maps the current landscape and identifies where opportunity is concentrated. For sports administrators and athletes, it contextualises how material technology is reshaping competitive performance across disciplines.

The report does not advocate for carbon fibre as a solution to all challenges facing Indian sport. It does, however, argue that ignoring the material revolution underway in global sports equipment manufacturing carries real costs in competitiveness, in export potential, and in the equipment available to the next generation of Indian athletes.

1.4 Research Methodology and Data Sources

This report draws on a multi-source, mixed-methods research framework combining secondary literature review, quantitative market data, policy document analysis, and industry intelligence. Given the cross-sectoral scope of the study spanning sports science, advanced materials engineering, trade economics, and public policy, no single data source provides sufficient coverage. The methodology is therefore intentionally pluralistic, prioritising corroboration across independent sources over reliance on any single dataset.

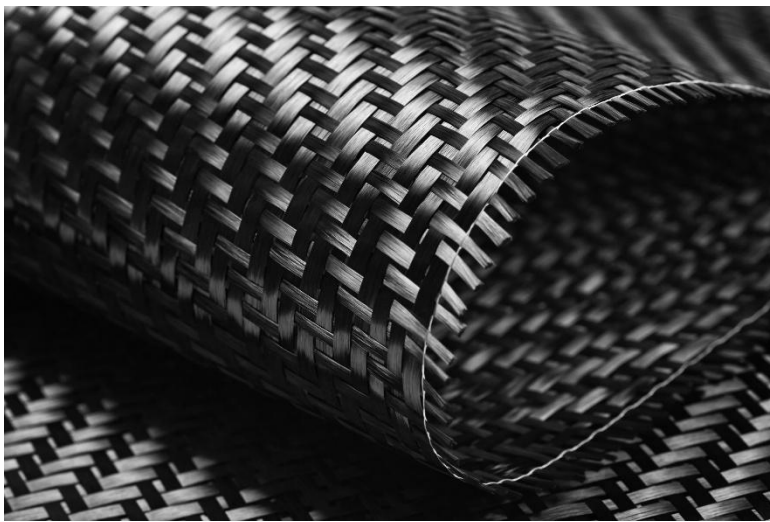
1.5 Key Limitations

This report relies entirely on secondary data. No primary field surveys, structured interviews with Indian athletes or manufacturers, or direct plant visits were conducted during the study period. Findings on domestic manufacturing capability are therefore based on publicly available company disclosures and media reporting, which may not fully reflect current ground-level conditions.

Chapter 2: Understanding Carbon Fibre

2.1 What is Carbon Fibre? Properties and Classification

Carbon fibre is an incredibly strong, lightweight, and versatile advanced material composed of microscopic filaments measuring just 5 to 10 micrometres in diameter, roughly a fraction of the thickness of a human hair. At the atomic level, its structure closely resembles that of graphite, consisting of sheets of carbon atoms arranged in a regular two-dimensional hexagonal lattice. Because these crystals align perfectly parallel to the fibre's long axis, the material exhibits a legendary strength-to-weight ratio. Carbon fibre has a low density of approximately 1.6-2.0 g/cm³, making it significantly



Source: SMI Composites

lighter than aluminium, which has a density of 2.7 g/cm³. Despite this extreme lightness, it offers exceptional mechanical properties, including a tensile strength of up to 7 GPa (approximately 500,000 psi) and a modulus of up to 900 GPa. This translates to a material approximately four times stronger than aluminium and 75% lighter than steel. Carbon fibre further exhibits excellent creep resistance,

minimal thermal expansion, and strong chemical and fatigue resistance.

2.1.1 Precursor Materials

The production of carbon fibre begins with organic polymers known as precursors. Polyacrylonitrile (PAN) is the undisputed industry standard, accounting for over 90% of all commercial carbon fibre production owing to its high carbon yield and superior mechanical performance. Alternative precursors include pitch derived from petroleum or coal tar, which produces fibres with an extremely high modulus and thermal conductivity but relatively lower tensile strength, and cellulose or rayon, which was historically significant in early development but is rarely used today due to its exceptionally low carbon yield.

2.1.2 Classification of Carbon Fibres

As illustrated in the Figure given below, carbon fibres can be classified along several dimensions, including precursor type, mechanical properties, thermal treatment, manufacturing method, function, and application grade¹.

Figure 4: A. Classification Based on Precursor Fibre Material (Annexure)

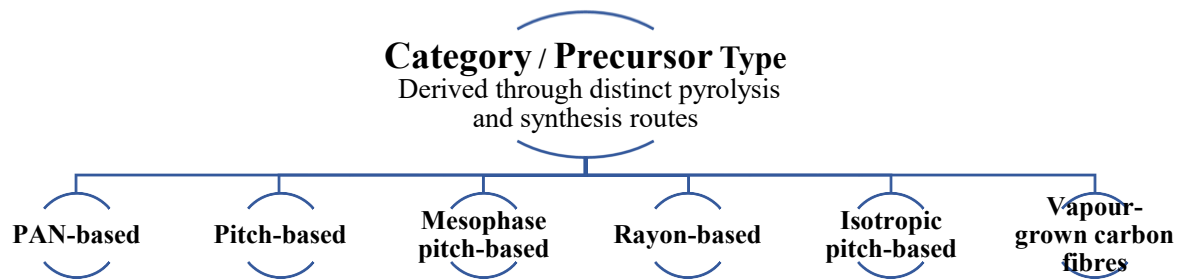
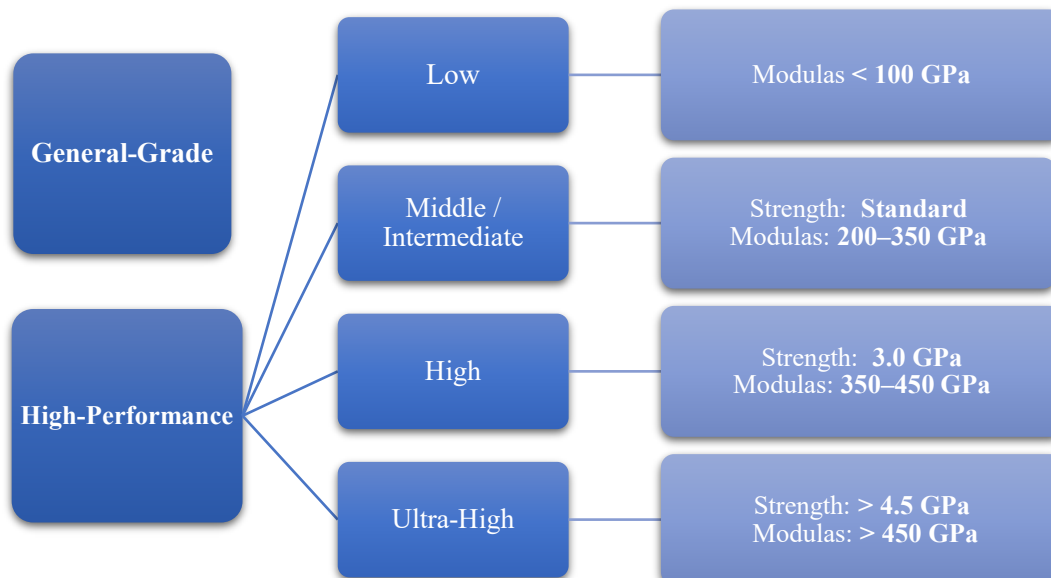


Figure 5: B. Classification Based on Mechanical Properties



¹ Rajak, D. K., Wagh, P. H., & Linul, E. (2021). Manufacturing Technologies of Carbon/Glass Fiber-Reinforced Polymer Composites and Their Properties: A Review. *Polymers*, 13(21), 3721. <https://doi.org/10.3390/polym13213721>

Figure 6: C. Classification Based on Final Heat Treatment Temperature (FHTT)

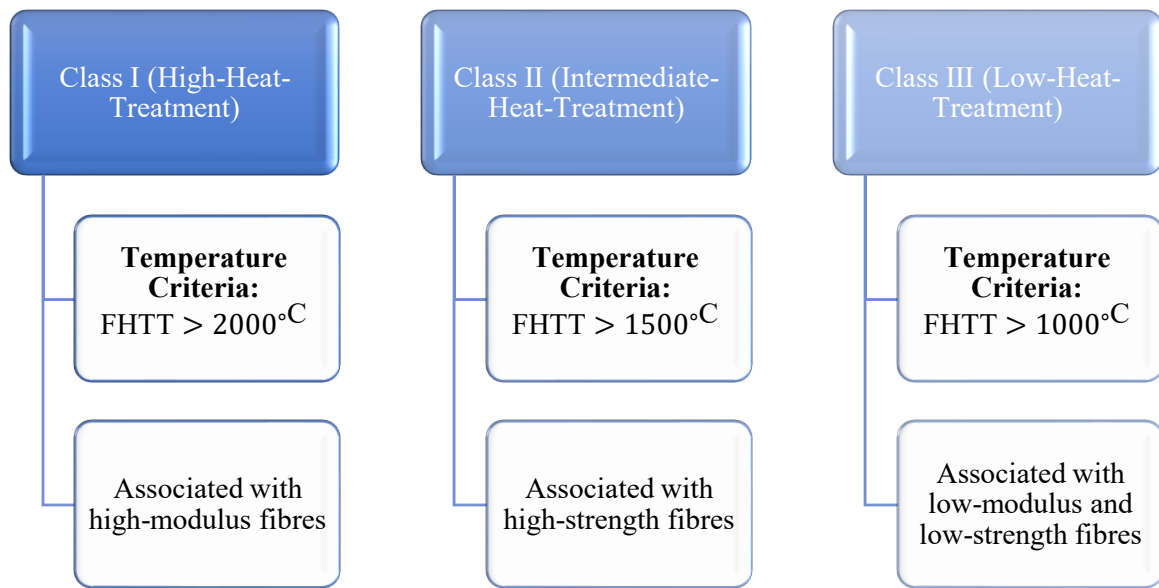


Figure 7: D. Classification Based on Manufacturing Method

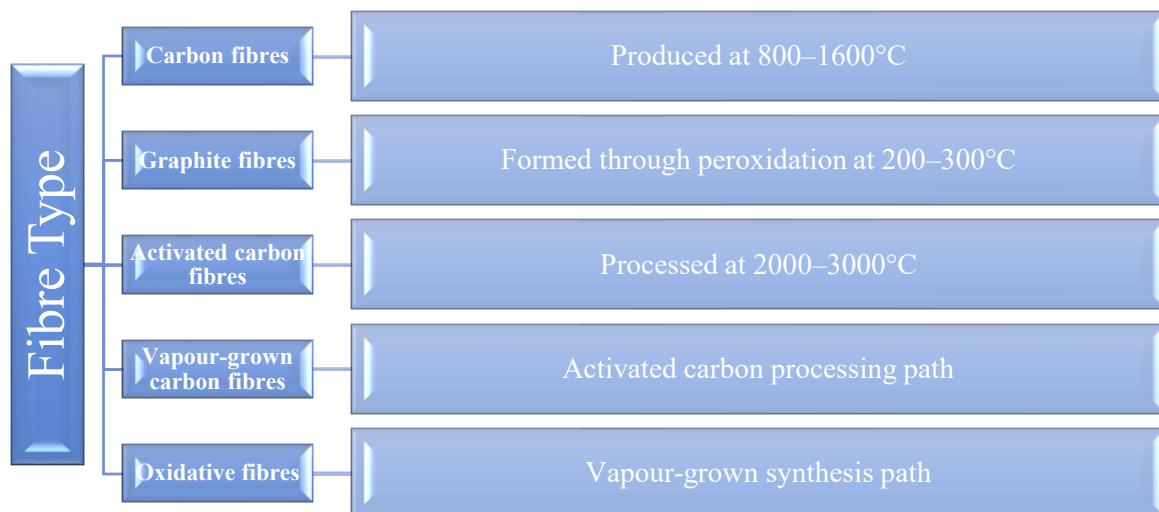
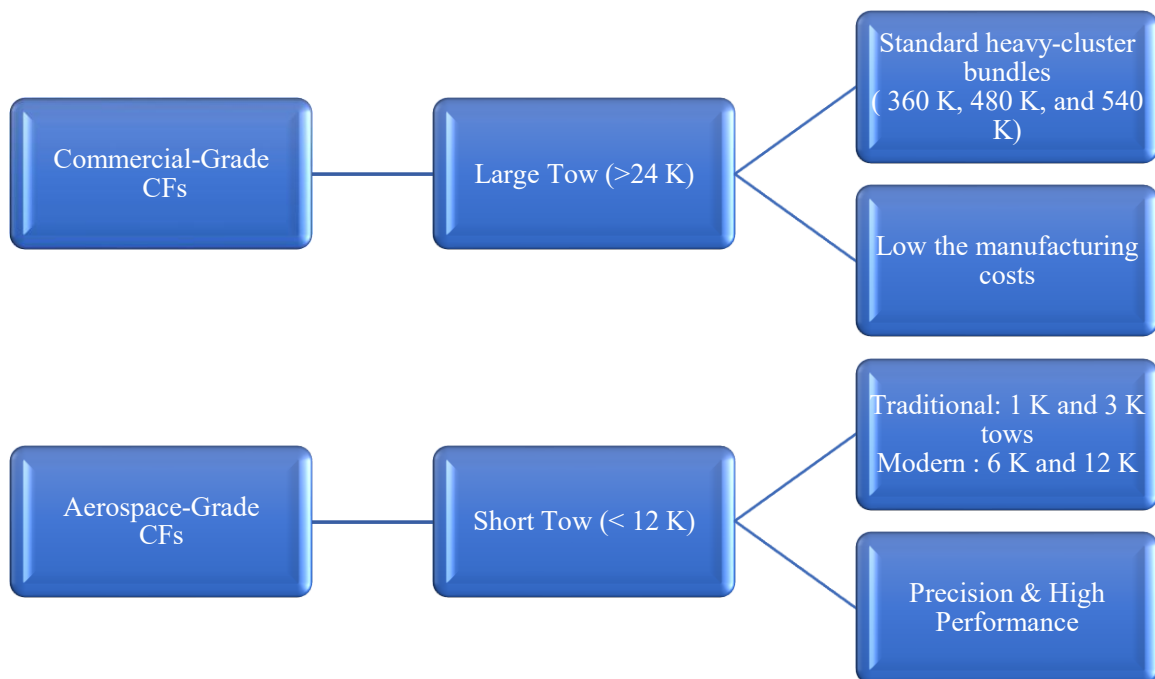


Table 3: E. Classification Based on Function

Category / Type	Core Technical Characteristics & Functions
Flame-resistant CFs	High thermal stability; formulated to resist ignition and tolerate extreme temperatures.
Load structure CFs	High tensile strength and modulus; optimised for load-bearing applications.
Activated CFs	High porosity and specific surface area; designed explicitly for adsorption activity.
Lubrication CFs	Low friction coefficient and self-lubricating properties; minimises mechanical drag.
Conductive CFs	High electrical/electronic conductivity; useful for grounding and EMI shielding.
Corrosion-resistant CFs	High chemical inertness; resists degradation from acids, bases, and harsh solvents.
Wear-resistant CFs	Exceptional surface durability and low wear rate under high friction conditions.

Table 4: F. Classification Based on Application Field²

² 1 K = 1,000 monofilament threads.

Short-tow (<12 K) denotes high-precision, low-volume aerospace variants;

Large-tow (>24 K) denotes high-volume, cost-optimised industrial bundles (e.g., 360 K–540 K) designed to reduce manufacturing overhead.

2.2 Carbon Fibre vs. Traditional Sports Materials and Other Fibres

For generations, wood was the foundational material for equipment such as hockey sticks and tennis rackets because of its natural flex and familiar feel. However, wood suffers from significant performance drawbacks; its natural grain creates inherent variations that lead to inconsistent performance, and it is highly susceptible to warping from moisture and breaking under high-impact stresses. To counter these issues, the industry initially shifted toward metals, with steel and aluminium becoming the standards for durability. While steel provides exceptional tensile strength, it is a highly dense and heavy material, which severely limits swing speeds and athletic agility. Aluminium emerged as a preferred alternative because it deforms under extreme stress rather than snapping, allowing it to absorb sudden impacts efficiently. Despite these advantages, metals possess uniform, isotropic properties, meaning their strength and stiffness are identical in all directions, which prevents engineers from customising the flex of the equipment. Furthermore, aluminium transmits high-frequency shock waves and vibrations upon impact, which can travel directly into an athlete's limbs and cause joint fatigue or repetitive strain injuries, such as tennis elbow.

Carbon fibre fundamentally disrupted this landscape by offering mechanical properties that metals cannot match. With a density of approximately 1.7 g/cm³, carbon fibre is roughly 40% lighter than aluminium (which has a density of ~2.7 g/cm³) (Table). Structurally, carbon fibre composites can offer up to five times the strength of standard grade steel while remaining five times lighter, enabling manufacturers to build equipment with strength-to-weight ratios that are 4 to 10 times higher than those of traditional metals. Most importantly, carbon fibre is anisotropic, meaning its structural properties are highly directional³. By strategically altering the orientation of the carbon weaves during the layup process, engineers can design a bicycle frame or a racket that is extremely stiff in one specific plane but flexible and compliant in another.

Table 5: Carbon Fiber Epoxy Matrix vs. 6061-T6 Aluminium Mechanical Properties

Material	Density (g/cm ³)	Tensile Strength (MPa)	Tensile Modulus (GPa)	Elongation at Break (%)
Carbon Fiber	1.79	569	291	2
6061 Aluminium	2.7	310	68.9	12

Source: Zhao, J. (2023). Carbon fiber comparison to traditional materials for the automotive industry. *Applied and Computational Engineering*, 25, 181-185. <https://doi.org/10.54254/2753-2721/25/20230756>

This structural customisability allows for profound enhancements in athletic performance and biomechanical efficiency. In stick-and-bat sports, the tailored flex of carbon fibre creates precise 'kick

³ In carbon fibre, the individual carbon atoms are linked together with a crystalline structure oriented along the direction of the fibres. The atoms slip over one another or distort in relation to one another more easily in some directions than others. This directionality of properties is what makes carbon fibre anisotropic.

points' that store and explosively release kinetic energy, generating faster shot velocities and a powerful 'trampoline effect' that wooden and aluminium counterparts cannot replicate. In cycling and racket sports, carbon fibre's unique structure acts as a natural dampener, effectively absorbing harsh impact vibrations before they reach the athlete's joints. This inherent shock absorption drastically reduces player fatigue, improves comfort, and allows athletes to maintain peak stamina over extended periods of play.

However, the transition to carbon fibre does introduce significant trade-offs, primarily regarding failure modes and cost. Because carbon fibre lacks the ductility of aluminium, it is inherently brittle; when it is overloaded or subjected to a sharp, sudden impact, the material does not bend or dent, but rather cracks, delaminates, or shatters catastrophically. Additionally, producing carbon fibre components requires extreme temperatures, specialised moulds, and skilled labour, making the material substantially more expensive, often 5 to 10 times the cost per pound compared to aluminium. Despite these challenges, the unparalleled advantages in speed, power transfer, and weight reduction have cemented carbon fibre as the definitive standard for high-performance sports equipment. Kobayashi et al. (2026) conducted a meta-analysis of 14 trials confirming that carbon-plated running shoes reduce metabolic demand. (Annexure 1)

2.3 Key Manufacturing Processes

The manufacturing of carbon fibre is a highly controlled, multi-stage process that involves converting an organic polymer precursor into a high-strength, carbon-rich filament. While precursor materials such as cellulose (rayon) and petroleum pitch are used, polyacrylonitrile (PAN) remains the industry standard, accounting for approximately 90% of all commercial carbon fibre production.

2.3.1 PAN-Based Carbon Fibre Production

PAN-based carbon fibre is produced through the following sequential stages:

Polymerisation: The process begins by converting acrylonitrile monomers into long polymer chains. Small amounts of comonomers (2–10%), such as methyl acrylate or itaconic acid, are added to improve fibre performance and aid subsequent thermal processing.

Fibre Spinning: The polymerised PAN is spun into continuous filaments using wet spinning or dry-jet wet spinning techniques. The polymer is extruded through a spinneret into a coagulation bath, after which the fibres are washed and mechanically stretched to align the polymer molecules parallel to the fibre axis, a step critical to enhancing final mechanical strength.

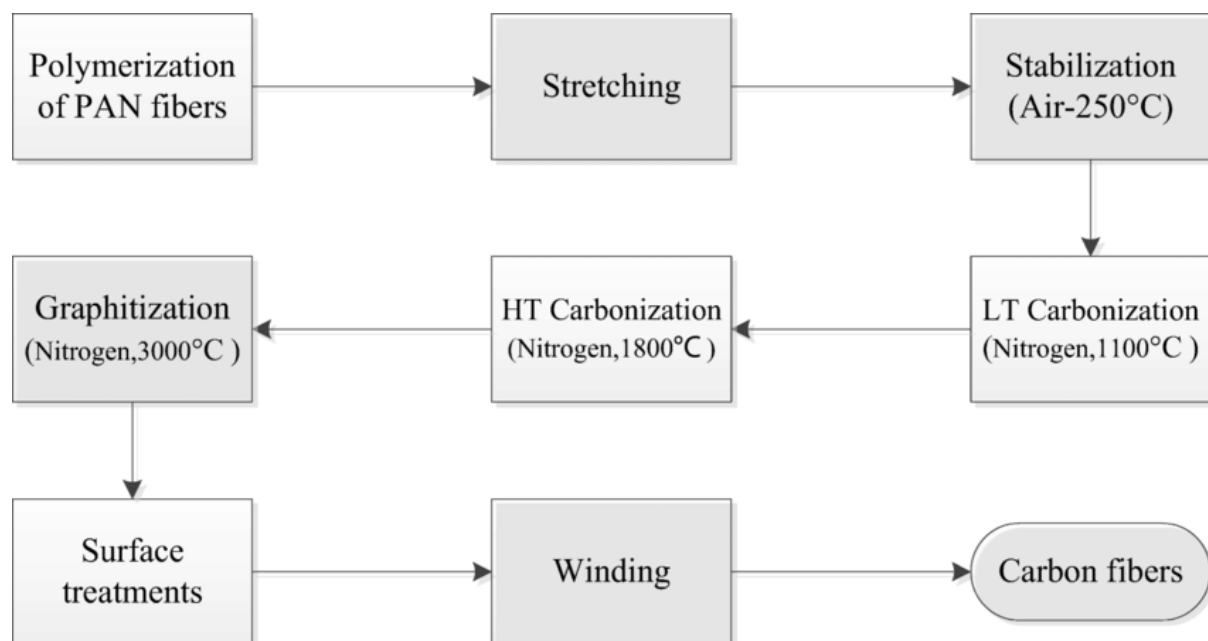
Thermal Stabilisation (Oxidation): The spun fibres are heated in air at 200–300°C for several hours. This oxidation process induces chemical cyclisation, transforming the linear polymer chains into thermally stable ring structures that will not melt during subsequent high-temperature treatment.

Carbonisation: The stabilised fibres are transferred into a high-temperature furnace operating in an inert atmosphere (nitrogen or argon) at 1000–2000°C. Almost all non-carbon elements, hydrogen, nitrogen, and oxygen are expelled as gases, leaving behind a tightly bonded carbon filament of 93–95% purity.

Graphitisation (Optional): For applications requiring exceptionally high stiffness, the fibres undergo an additional heating phase at 2500–3000°C. This significantly improves the alignment of internal graphite-like crystal layers, boosting the elastic modulus of the fibre.

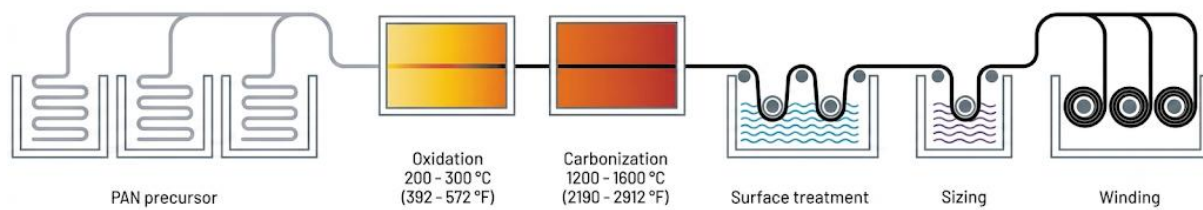
Surface Treatment and Sizing: Since pure carbon fibre does not naturally bond well with polymer resins, the fibres undergo mild electrochemical anodic oxidation, introducing polar functional groups and microscopic surface texture to promote adhesion. They are then coated with a protective sizing agent typically epoxy or polyvinyl alcohol before being wound onto bobbins for shipment and fabrication.

Figure 8: Carbon fibre manufacturing process steps



Source: A multi-objective Gaussian process approach for optimization and prediction of carbonization process in carbon fibre production under uncertainty - Scientific Figure on ResearchGate. Available from: https://www.researchgate.net/figure/Carbon-fibre-manufacturing-process-steps-adapted-from-12_fig3_333786237 [accessed 10 Jun 2026]

Figure 9: Overview of Carbon Fibre production



First, they are oxidised at 200 to 300 degrees Celsius and then carbonised at 1200 to 1800 degrees Celsius. What remains are threads with a very high carbon content and high strength. After surface treatment and application of a sizing, the carbon fibre is wound up and is ready for use.

source: <https://www.sglcarbon.com/en/carbon-fibres-and-cfrp/>

2.3.2 Composite Fabrication Techniques

Raw carbon fibre on its own is just a collection of threads. To become a usable product a bicycle frame, a hockey stick, or a racket needs to be combined with a **resin** (usually epoxy) and shaped. There are a few ways this is done:

Open Moulding (Hand Layup)- A traditional, cost-effective method where carbon fibre fabric layers are manually placed into a one-sided mould and resin is applied by hand. While widely used, it is relatively slow and labour-intensive, and is progressively being replaced by more precise techniques

Prepreg Layup and Autoclave Curing -This process uses prepregs, carbon fibre sheets pre-impregnated with a precisely controlled resin ratio. The prepreg layers are placed onto a mould and cured inside an autoclave at 6–7 bar pressure and 120–180°C. This eliminates voids and delivers superior mechanical properties, making it the standard in aerospace manufacturing.

Resin Transfer Moulding (RTM)- Dry carbon fibre preforms are placed inside a closed mould, and low-viscosity resin is injected under pressure until the cavity is fully filled. RTM is well-suited to intricate geometries and medium-to-high volume production.

Vacuum Infusion / VARTM -Vacuum pressure is used to draw resin through dry carbon fibre layers, removing air bubbles and reducing void content. This cost-effective method is widely used for large structures such as wind turbine blades.

Filament Winding- Continuous carbon fibre tows are impregnated with resin and wound around a rotating mandrel in a precisely controlled pattern. The process is highly automated and is primarily used for hollow cylindrical structures such as pressure vessels and pipes.

Pultrusion- Continuous fibres are pulled through a resin bath and a heated die that simultaneously shapes and cures the material, producing continuous composite profiles with a constant cross-section, such as beams and rods (Figure)

Figure 10: CFTF tows mounted to spindles and traversing guides for pultrusion

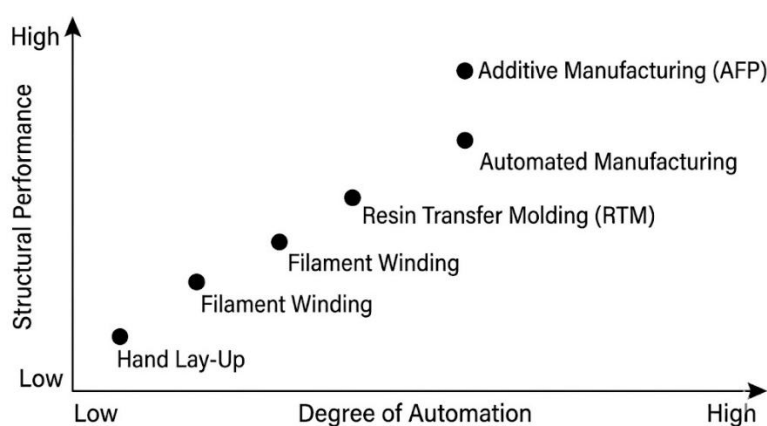


Source: (Ennis et al., 2019, p. 13, Figure 1-1)

Automated Fibre Placement (AFP) and Automated Tape Laying (ATL) Robotic systems precisely place narrow prepreg tapes or tows onto complex mould surfaces at high speed, offering accuracy, repeatability, and cycle time reductions of up to 70%. AFP and ATL are increasingly viable for high-volume automotive and aerospace production.

Additive Manufacturing (FDM / 3D Printing) Short carbon fibres are used to reinforce plastic filaments, which are extruded layer-by-layer to build three-dimensional objects. This approach enables rapid prototyping and the fabrication of complex geometries that would be difficult to achieve through conventional methods.

Figure 11: Schematic comparison of composite manufacturing methods showing degree of automation versus structural performance



Source: Mamat, R., Rashid, M. I. M., Syahir, A. Z., Erdiwansyah, Yusop, A. F., & Tamimi, A. (2025). Carbon fibre for applications in aerospace: A review. *Journal of Alloys and Metallurgical Systems*, 12, Article 100227. <https://doi.org/10.1016/j.jalms.2025.100227>

Chapter 3: Carbon Fibre Across Sports Disciplines through Global Evidence

3.1 Cycling

3.1.1 Historical Progression

The application of carbon fibre in cycling represents one of the earliest and most transformative shifts in sports equipment history. In 1970, Gerald O'Donovan of Carlton Cycles began experimenting with carbon composites, leading to the unveiling of the first carbon fibre bicycle at the Harrogate Cycle and Motorcycle Show in 1971. By 1976, the U.S. Olympic team was competing on the 'Graftek G-1', an early hybrid frame featuring carbon fibre tubes bonded to aluminium lugs (see Image).

Figure 12: The article details the unveiling of a historic 13 lb. 12 oz. prototype carbon-fibre track bicycle one of the earliest recorded examples in cycling history.



Date of Publication: Circa July / August 1971

However, these early generations relied on tube-to-tube bonding methods that limited design flexibility and occasionally resulted in fragile, inconsistent joints prone to failure. A major engineering leap occurred in the 1980s. The Specialized Epic, introduced in 1983, became the first mass-produced carbon frame, followed by the Kestrel 4000 in 1986, which pioneered the first true 'monocoque' (single-piece

moulded) carbon bicycle frame. By the late 1980s and 1990s, brands like Giant and Trek introduced CAD-designed and moulded carbon frames, transitioning the material from a rare novelty to the definitive standard in professional cycling.

3.1.2 Frame Design, Aerodynamics, and Weight Optimisation

Modern carbon fibre frames dominate elite cycling due to the material's unique anisotropic nature, which allows engineers to precisely tune a frame's directional stiffness and compliance. Unlike isotropic metals such as aluminium or steel, which exhibit uniform properties in all directions, carbon fibre layups can be strategically oriented. For instance, a frame can be engineered to be laterally and torsionally stiff around the bottom bracket, ensuring that pedalling energy is transferred directly into forward motion without power loss while remaining vertically compliant to absorb high-frequency road chatter.

To achieve the ultimate balance of weight, aerodynamics, and comfort, manufacturers meticulously blend different grades of carbon fibre within a single frame. Not all carbon fibres are the same. In cycling, Today's T700, T800, and T1000 fibres are among the most widely used (Annexure 1). Resilient T700 fibres are often used to enhance comfort in endurance models, stiffer T800 fibres are utilised to shed weight, and ultra-high-modulus T1000 fibres are placed in maximum-stress zones to deliver explosive power transfer. Coupled with monocoque construction, which enables the moulding of highly complex, drag-reducing aerodynamic shapes, these advancements have caused frame weights to drop dramatically, with some elite road frames now weighing less than 700 grams.

3.1.3 Carbon Fibre Wheels, Handlebars, and Components

Beyond the frame, carbon fibre is extensively used in ancillary cycling components to further optimise weight and biomechanical comfort. Because carbon fibre naturally dampens vibrations, utilising the material in contact points such as seat posts, handlebars, and forks significantly reduces the transmission of harsh road buzz to the rider's hands and spine. This reduction in vibration minimises joint fatigue and numbness, providing a noticeably 'smoother' ride over long distances compared to pure aluminium setups. Additionally, carbon fibre is utilized in the manufacturing of rigid cycling shoes, which protect the cyclist's feet and maintain optimal power transfer and balance.

Figure 13: Carbon Fibre Bicycle Handle Bar



Source: Juli FRP (Dongguan Juli Carbon Fibre Products Co., Ltd.)

3.1.4 Manufacturing Processes: Bladder Moulding and RTM in Cycling

The manufacturing of modern carbon bicycles has largely moved away from the early tube-to-tube bonding methods in favor of advanced automated and single-piece moulding techniques. Monocoque construction involves placing hundreds of precisely cut pre-impregnated carbon fibre sheets (prepreg) into a mould, which is then cured under controlled temperature and pressure conditions to form a seamless structure.

To improve efficiency and structural integrity, Resin Infusion processes such as Resin Transfer Moulding (RTM) and vacuum-assisted resin transfer moulding (VARTM) are increasingly utilized. These processes use vacuum pressure to draw liquid resin through dry carbon fibre layers placed inside a mould, ensuring complete wet-out of the fibres while eliminating internal air voids and excess resin. As the industry advances, manufacturers are also exploring automated layup systems utilizing robotics to improve precision, alongside the integration of new thermoplastic resins.

3.1.5 Regulatory Limits

the Union Cycliste Internationale (UCI) plays a critical regulatory role in the modern application of carbon fibre, particularly concerning environmental sustainability. Recognising that the sports industry is the third-largest global user of carbon fibre, and that most high-end carbon sports equipment has an average lifespan of only three years before ending up in landfills, the UCI has partnered with the Carbon Fibre Circular Alliance. The UCI is actively driving initiatives to reclaim and realign broken carbon fibre components into continuous unidirectional tapes (AFFT tape) using HiPerDiF technology, with the regulatory ambition of showcasing fully recycled carbon equipment at future Olympic Games.

3.2 Racket Sports

The global market for racket sports is experiencing strong growth across tennis, badminton, and squash, driven mostly by increasing player participation, innovative equipment technology, and the rapid expansion of online retail platforms. Within this landscape, the global carbon fibre badminton racquet sector reached a valuation of \$1.8 billion in 2025 and is anticipated to climb to \$3.2 billion by 2034, growing at a CAGR of 6.6%. This specific segment relies heavily on the Asia-Pacific region, which led the industry in 2025 with an \$876.6 million valuation representing a 48.7% share of global revenue and is projected to sustain its dominance through 2034 with an even higher regional CAGR of 7.2%.

Meanwhile, the market for premium tennis gear is posting consistent upward gains. Data from HTF Market Intelligence indicates that the global carbon fibre tennis racket market, fueled by player demand for high-performance materials, sits at a current value of \$1.2 billion and is on track to expand by 5.9% annually to touch \$1.8 billion by 2033. In addition to these more established sectors, the global squash rackets market is generating notable momentum, climbing from a baseline value of \$243.1 million in 2023 with a projected CAGR of 5.3% running through 2030. While rising interest in both recreational and competitive play has continuously backed squash gear sales, the sector is set for a historic milestone and unprecedented demand after the International Olympic Committee confirmed squash's official debut at the 2028 Los Angeles Olympic Games, establishing sustained interest in premium sports merchandise.

3.2.1 The Transition from Aluminium and Steel to Carbon Fibre Frames

Historically, aluminium replaced wood due to its improved durability, but aluminium frames are susceptible to permanent bending or deformation under the extreme stress of high-impact volleys. Carbon fibre revolutionised this landscape by offering a significantly lower density, allowing manufacturers to create rackets that are up to 30% lighter than their aluminium counterparts without compromising structural integrity. The high tensile strength of carbon fibre ensures that the racket maintains its carefully engineered shape and sweet spot, providing players with higher swing speeds and reduced fatigue during extended matches (Table).

Table 6: Material Comparison

Feature	Carbon Fibre	Aluminium	Steel
Weight	Very Light	Moderate	Heavy
Durability	High	Medium	High
Flexibility	Controlled Flex	Low	Very Low
Shock Absorption	Excellent	Moderate	Low
String Tension Support	High (24–32 lbs)	Low	Medium

Source: <https://www.sparkshot.com/news-show-1034426.html>

3.2.2 Stiffness, Balance, and Vibration Damping in Racket Design

The integration of carbon fibre in racket design allows engineers to meticulously tune the stiffness, balance, and vibration damping of the equipment. Stiffness, typically measured on a scale between 55 and 72, dictates energy transfer; a stiffer racket transfers greater impact energy to the ball for more power but also transmits higher impulse forces to the player's arm. However, carbon fibre possesses unique molecular properties that excel at vibration dampening. The interwoven carbon filaments effectively absorb and dissipate high-frequency shock waves before they reach the athlete's wrist and elbow, significantly lowering the risk of repetitive stress injuries such as tennis elbow. Advanced structural innovations, such as triple-core carbon handle designs, have been shown to reduce initial impact shock by at least 65% and demonstrate substantially faster vibration-damping times compared to traditional hollow-core designs.

3.3 Badminton

A recent survey conducted across 21 countries showed that 44% of youth between the ages of 15 and 24 now prefer badminton over other racquet sports due to its agility and indoor nature. The rise of badminton leagues such as the Premier Badminton League (PBL) in India and the Chinese Badminton Super League has boosted demand for carbon composite rackets, which currently make up 59% of racket sales globally. In badminton, carbon fibre technology has driven an economic and technological boom by enabling the creation of ultra-lightweight rackets. Lightweight frames, defined as having unstrung weights between 75 and 84 grams, now dominate the industry, capturing 62.4% of the market share. Pushing the weight below the 80-gram threshold while maintaining the rigidity necessary for explosive shuttle repulsion presents a significant engineering challenge. Leading brands have overcome this by utilizing proprietary high-modulus and ultra-high-modulus carbon fibre grades, often reinforced with nano-carbon technologies. These advanced materials allow for thinner frame walls without sacrificing torsional stiffness, granting players lightning-fast reflexes, quicker stroke preparation, and tighter net play. This reduction in weight has not only elevated professional play but has also spurred significant participation growth among women and youth demographics, who benefit biomechanically from the highly responsive, sub-80-gram frames.

3.4 Tennis

Carbon fibre has enabled unprecedented levels of player customization in tennis through the manipulation of variable flex ratings. By adjusting the carbon fibre layups, specifically the orientation, thickness, and types of weaves used, manufacturers can create rackets that cater to highly individualized playing styles. A stiffer frame minimizes energy loss during the loading and unloading phase of a swing, generating maximum power for aggressive players. Conversely, more compliant layups create a wider hysteresis curve, absorbing more energy but offering greater control and a softer feel. This strategic layering enables players to select rackets ranging from 220-gram lightweight entry models to 370-gram

stiff professional frames, optimizing the balance between raw power, directional control, and physical comfort.

Table 7: Comparison of carbon fibre to other common tennis racket materials

Material	Strength-to-Weight Ratio	Stiffness	Vibration Damping	Power Potential
Carbon Fibre	Excellent	High	Very Good	High
Aluminium	Fair	Low	Poor	Medium
Titanium	Good	Medium	Fair	Medium
Wood	Poor	Very Low	Good	Low

Source: Mantis Sport. Carbon fibre tennis rackets. <https://mantissport.com/blogs/news/carbon-fibre-tennis-rackets>

3.5 Squash

Squash, an exceptionally fast-paced indoor sport played with a smaller ball, demands rapid changes in direction and instantaneous reaction times. The introduction of carbon fibre completely phased out heavy wooden frames, pushing the weight of modern squash rackets down to as little as 100 grams. This extreme lightness allows players to react faster, swing harder, and tire less easily during relentless rallies. Additionally, the immense tensile strength of carbon fibre enabled engineers to drastically increase the surface area of the racket face, providing a much larger and more forgiving sweet spot. However, the use of carbon fibre in constrained, walled environments like a squash court introduces a trade-off between peak performance and durability.

3.6 Cricket

3.6.1 Carbon Fibre in Protective Gear

While the cricket bat's blade is strictly bound by historical regulations, the sport's protective equipment faces no such material constraints. As the speed and physical intensity of modern cricket have increased, the need for lightweight, high-impact resistance gear has become paramount. Leading manufacturing clusters in the Indian subcontinent, such as Jalandhar and Meerut in India, and Sialkot in Pakistan, produce millions of units of protective gear annually, including batting gloves, wicket-keeping pads, thigh guards, chest guards, and helmets. To protect batsmen from the severe trauma of a hard leather ball travelling at speeds exceeding 140 km/h, manufacturers are increasingly looking toward advanced composites. Carbon fibre and other high-strength polymers are integrated into the outer shells of helmets and the inserts of gloves and pads. These materials provide critical vibration dampening and impact absorption without adding restrictive bulk, allowing batsmen to remain agile and heavily protected simultaneously.

3.6.2 Bat Handles and Composite Cores

Although the blade of the bat is restricted to wood, the handle has historically been a site of significant engineering innovation. The transition began in the 1830s when the advent of faster 'round-arm' bowling caused one-piece wooden bats to shatter frequently, leading carpenters to introduce a separate, spliced handle made of flexible cane and ash. In the 21st century, manufacturers began experimenting with advanced composites to further enhance power and durability. In 2005, Kookaburra released a highly controversial bat featuring a carbon fibre-reinforced polymer support running down the spine of the blade. Famously championed by Australian captain Ricky Ponting, this carbon fibre spine prolonged the life of the bat and provided immense structural support. Shortly after, in 2006, Newbery introduced a fully carbon fibre composite handle, and Gray Nicolls followed suit with the 'Fusion' bat in 2007. These hybrid designs aimed to maximize the energy transfer from the batsman's hands through the willow, generating unprecedented hitting power.

3.7 Hockey Carbon Fibre Sticks and India's Olympic Legacy

3.7.1 Evolution of Hockey Stick Materials

The evolution of the hockey stick is a testament to the relentless pursuit of athletic performance. Historically, sticks were crafted from woods like ash, maple, and birch, which offered a natural, satisfying feel but suffered from inconsistent flex profiles and a high susceptibility to moisture warping and sudden breakage. The industry witnessed a transitional phase in the 1990s with the introduction of aluminium shafts paired with composite blades, but the definitive turning point occurred in the early 2000s with the advent of the fully composite, one-piece stick. A major catalyst for this global transition was the FIH (Fédération Internationale de Hockey) decision in 2001 to officially approve composite hockey sticks for international events, a regulatory shift that effectively forced traditional wood manufacturers across the globe to rapidly adapt to composite technologies or face obsolescence.

3.7.2 Power, Accuracy, and Feel

The integration of carbon fibre fundamentally alters the biomechanics of a hockey shot, primarily through the precise manipulation of power, accuracy, and sensory feel. Unlike early models that utilized unidirectional carbon sheets prone to cracking under stress, modern high-performance sticks employ multi-directional woven carbon layups that distribute impact forces evenly, improving structural integrity by up to 30%. This extreme stiffness ensures that minimal energy is lost in the flexing of the stick; instead, kinetic energy is stored and explosively transferred directly into the ball or puck during hits, slaps, and drag flicks. Engineers customize these energy transfers using specific 'kick points' such as low-kick designs for lightning-fast snap releases, and mid-kick profiles for maximum power generation. However, because pure 100% carbon fibre can feel overly rigid and unforgiving,

manufacturers frequently blend carbon with fibreglass and Kevlar to fine-tune the stick's flexibility. This allows them to offer 'playmakers' a softer touch for superior ball control and 3D skills, while reserving ultra-stiff pure carbon models for dedicated power hitters.

Table 8: Comparing Stick Materials: Wood vs. Aluminium vs. Composite

Material	Weight	Durability	Cost	Performance
Wood	Heaviest	Prone to cracking and moisture damage over time	Most affordable / budget-friendly	Solid traditional feel, but lacks the modern snap and pop
Aluminium	Medium	Highly durable and rigid, but can suffer from permanent bending or lack of flex	Moderate	A noticeable lack of flexibility makes it less popular for modern performance needs
Composite	Lightest	High resistance to cracking, structural wear, and moisture	Premium / highest price point	Dominates in shot speed, precision, and flex customisation

Source: Information adapted from [Pro Stock Hockey's Guide on Hockey Stick Materials](#).

3.7.3 Brands, Standards, and Benchmarks

Global hockey stick manufacturing has evolved into a highly sophisticated, technology-driven industry dominated by major international brands such as Bauer, CCM, Warrior, True, and Sherwood. These industry leaders rely on ultra-high-modulus aerospace-grade carbon fibres, such as T800 and T1100G, which provide maximum pop and responsiveness while shedding unprecedented amounts of weight. The performance of these sticks is equally dependent on advanced, proprietary resin systems such as Bauer's MonoComp, CCM's Nanolite Carbon Layering, and True's RESFLO, which aggressively bond the carbon layers, reduce micro-cracking, and maximize impact resistance. Furthermore, construction methods have shifted toward true one-piece moulding, where the shaft and hollow-core, foam-injected blade are seamlessly integrated. This highly calculated layer placement establishes a global benchmark for consistency, energy transfer, and highly tuned torsional rigidity.

3.7.4 Indian Hockey's Equipment Gap

Despite India's profound historical legacy in hockey, highlighted by its unmatched record of Olympic gold medals, the nation currently faces a significant equipment gap driven by structural manufacturing constraints. While elite Indian athletes have access to the imported, high-performance composite sticks utilized on the global stage, grassroots players often rely on inferior or outdated equipment. The core of this issue lies in the domestic manufacturing landscape, which is heavily concentrated in Micro, Small, and Medium Enterprises (MSMEs) located in traditional hubs like Jalandhar and Meerut. The cluster in Meerut has over 35,200 registered units compared to 3,000 in Jalandhar. These legacy clusters, which once dominated the global wooden stick market, have struggled to fully scale into advanced

composite manufacturing due to limited access to modern technology and the high costs of imported raw materials. Because India does not yet produce commercial raw carbon fibre, domestic manufacturers must import these advanced polymers and face steep customs duties. This significantly erodes their profit margins, inflating the cost of high-quality gear and creating a stark divide between elite access and grassroots affordability.

3.7.5 Domestic Manufacturing Potential for Carbon Fibre Hockey Sticks

India is actively mobilizing to bridge this technological gap and unlock its domestic manufacturing potential for advanced carbon fibre hockey sticks. Pioneering firms like Arvind Composites are already demonstrating domestic capabilities, manufacturing high-quality hockey sticks utilizing 24T carbon and high-strength E-Glass within state-of-the-art facilities equipped with robotic and CNC technologies. This domestic potential is poised for a massive acceleration following the Union Government's commitment to begin indigenous raw carbon fibre production by 2025–26, supported by 156 research projects sanctioned under the National Technical Textiles Mission. By eliminating the reliance on expensive imported carbon fibre and overcoming structural supply chain hurdles, Indian MSMEs will be uniquely positioned to manufacture globally competitive, affordable composite hockey sticks. This shift promises to empower a new generation of athletes and reclaim India's dominance in the global high-tech sports economy.

3.8 Athletics and Winter Sports Poles, Blades, and Protective Gear

3.8.1 Pole Vault

The fundamental mechanics of pole vaulting, specifically the storage and explosive release of kinetic energy, have served as a foundational inspiration for advanced carbon fibre engineering across sports. Because carbon fibre boasts exceptional tensile strength and customisable flexibility, it functions as a highly efficient mechanical spring. This ability to capture downward or forward kinetic energy, store it momentarily within the flexed composite matrix, and aggressively return it to the athlete is the defining characteristic that enables elite performance in both vaulting and running disciplines.

3.8.2 Paralympic Blades

The development of the carbon fibre running blade represents one of the most profound biomechanical advancements in modern sports history. The first running blade, known as the 'Flex-Foot', was designed in the 1970s by Van Phillips, an American inventor and amputee who based his designs on the mechanics of diving boards, pole vaulting, and the movement of kangaroos and cheetahs. Modern running blades, such as the iconic 'Cheetah' manufactured by Össur, are constructed from 100% carbon fibre. These prostheses achieve their immense strength by fusing between 30 and 90 microscopic layers of carbon fibre, each thinner than a human hair, using highly precise, computer-controlled

manufacturing. The backwards-curving design mimics the hind leg of a big cat, acting as a spring that compresses to store the runner's energy upon impact and releases it to propel the athlete forward.

The integration of these blades has fundamentally transformed the Paralympic Games. Starting with the Sydney 2000 Games and accelerating through Tokyo 2020, carbon fibre prosthetics evolved from simple compensatory aids into highly customised, aerodynamic extensions of the athlete's body that maximise and sometimes exceed biological potential. However, the unprecedented efficiency and energy return of these teardrop-shaped blades have sparked intense ethical debates regarding 'technological doping,' as governing bodies continually struggle to balance technological innovation with the principles of fair and equitable competition.

3.8.3 Helmets and Protective Equipment Across Disciplines

In high-velocity sports such as cycling, skiing, and hockey, where the risk of traumatic brain and joint injury is severe, advanced composites and foams are critical for athlete safety. Carbon fibre is increasingly utilized in the rigid outer shells of helmets and protective padding because it is significantly stronger than traditional metals while adding virtually no restrictive bulk.

The primary function of this equipment is the efficient management of kinetic energy. Protective gear is engineered to absorb and distribute shock waves across the surface of the carbon composite shell, while inner layers of advanced foams with varying densities cushion the direct impact. For instance, modern helmet designs are incorporating hydrophilic porous materials that swell with water to drastically reduce deceleration rates during a crash, thereby minimising the risk of concussions.

3.8.4 Limitations of the Evidence

While laboratory data strongly support the structural benefits of carbon fibre, its translation to real-world performance is not absolute and is subject to significant biological and environmental variables. For instance, in running, biomechanical studies highlight that athletes are not uniformly responsive to carbon-plated shoes. In a running economy study, athletes with naturally high cadences (e.g., ~186 steps/min) and low vertical oscillation (under 8 cm) saw negligible VO₂ improvements (0% to 2.2%). Because these athletes act as 'shufflers' rather than 'bounders,' they fail to engage the carbon plate enough to realise its full energy-saving spring benefits.

Subjective perceptions of comfort and 'smoothness' can also be skewed by marketing and confirmation bias. In the cycling industry, while carbon frames are marketed for their vibration-dampening qualities, materials scientists and aerospace engineers argue that the compliance provided by tyre width and air pressure actually dominates road vibration absorption far more than the rigid carbon frame itself. Finally, assessing the real-world durability and catastrophic failure rates of carbon fibre remains challenging due to a lack of centralised reporting; in cycling, for example, crashes account for only an

estimated 6% of self-reported incidents, making it incredibly difficult to quantify how often carbon frames fail due to hidden fatigue or manufacturing defects rather than direct trauma.

Chapter 4: Status and Outlook of India's Sports Equipment Sector

The Indian sports economy reached ₹18,864 crore (\$2,134 million) in 2025, registering a 13.4% growth over ₹16,633 crore in 2024. The growth underscores not just expansion, but increasing structural maturity, with the sector nearly doubling in size from ₹9,530 crore in 2021, delivering an approximate CAGR of 18.6% over four years.

4.1 Global Market Share and Sector Overview

India's sports equipment manufacturing sector is at a pivotal crossroads, actively transitioning from a legacy of traditional, labour-intensive craftsmanship to a globally integrated ecosystem focused on high-tech advanced materials. While the global sports goods market was valued at approximately \$700 billion in 2024, with the sports equipment segment accounting for \$140 billion, India's share in global sports equipment exports remains marginal at just 0.5% (Annexure 1). 60% of Indian sporting goods are exported, and around 500,000 people are employed in the sports goods manufacturing sector in India. According to a Deloitte-Google report, the sports goods market was valued at US\$ 52 billion in 2024 and is expected to reach US\$ 130 billion by 2030, with a CAGR of 14%⁴.

4.1.1 Export Trend

Positioned as Asia's third-largest manufacturer, India has established a commanding presence in the global sports goods industry, effectively bolstering its international soft power by exporting a diverse portfolio ranging from cricket gear and fitness equipment to apparel and inflatable balls to more than 200 countries worldwide (Figure). The United States remains a primary destination, importing US\$ 137 million worth of Indian sports goods in FY25, and maintaining its momentum into the first quarter of FY26 (April–June) with US\$ 39 million in imports, followed by the UK at US\$ 12 million, Australia at US\$ 9 million, Germany at US\$ 6 million, and the Netherlands at US\$ 4 million. According to the India Exim Bank analysis, distinct regional preferences drive these numbers (Table).

Table 9: Regional Demand and Export Profiles

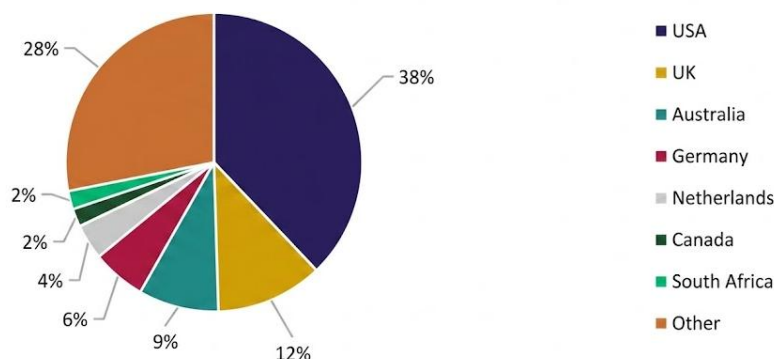
Target Market / Country	Leading Product Segments and Market Dynamics
United Kingdom (UK)	Leads global demand for sports balls (tennis and inflatable; excluding golf).
United States (US)	Dominates the export market for festive and carnival articles.

⁴ India Brand Equity Foundation. (2025, December). *Sports industry and export in India*. <https://www.ibef.org/exports/sports-industry-india>

United Arab Emirates (UAE)	Major importer of racket sports equipment (tennis/badminton) and sports balls.
Australia	Key secondary market for sports balls is following the UK.
Spain & Sri Lanka	Niche destinations are driving high demand for water sports equipment.
Latin America & Africa- (Brazil, Mexico, South Africa, etc.)	High-potential emerging markets are driving rapid demand for general athletic supplies.

Source: Singh, J., Raman, N., & Katiyar, P. (2021). *Indian sports goods industry: Strategies for tapping the export potential* (Working Paper No. 104). Export-Import Bank of India. <https://www.eximbankindia.in/sites/default/files/2025-07/143file.pdf>

Figure 14: Country-wise share of India's sporting goods exports (FY26*)



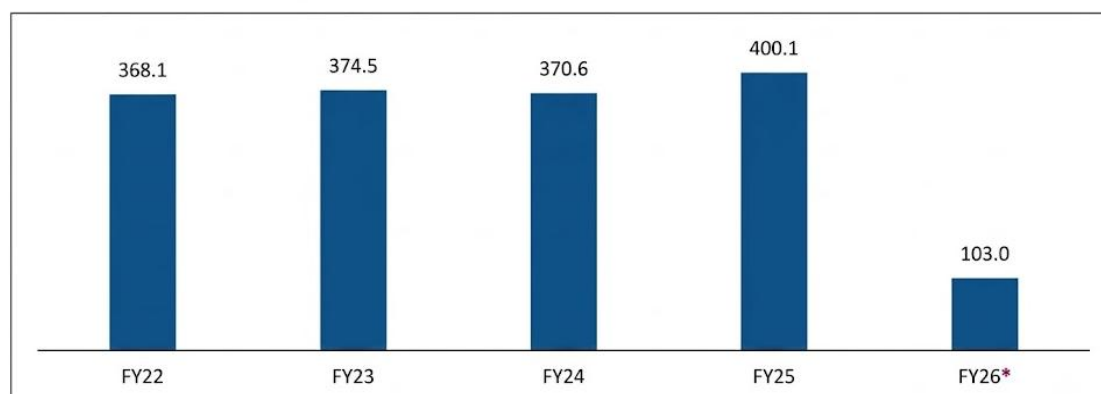
Source: Department of Commerce, Note: * - April-June

4.1.2 Export Commodity and Brand Partnerships

India's diverse sporting goods export sector encompasses a wide range of products, including toys, gymnasium equipment, athletic requisites, festive articles, and equipment for cricket, football, and racket sports. The industry has demonstrated steady financial growth, with total exports rising by 7.97% to reach US\$ 400.1 million in FY25 (Figure), followed by a strong performance of US\$ 102.97 million in the first quarter of FY26 alone. Driven by a major long-term competitiveness boost between 2010 and 2020, including a massive thirty-fold increase in racket sports exports, India has successfully established itself as a premier international manufacturing hub. Consequently, the country now serves as a key global sourcing destination for major international brands such as Wilson, Mitre, Lotto, and Umbro, particularly for inflatable balls and specialised athletic gear.

Figure 15: Exports Trend

India's sporting goods export trend (US\$ million)

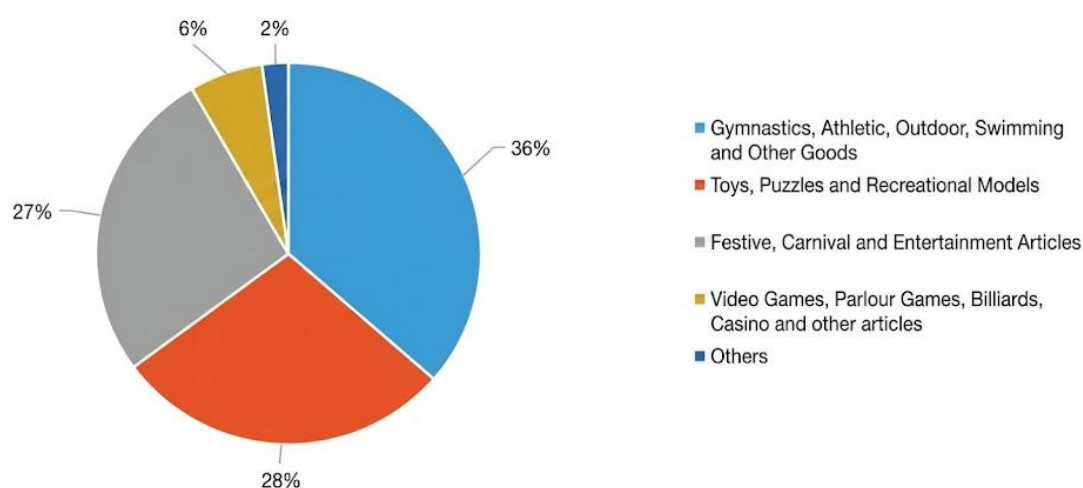


Source: Department of Commerce, Note: *- April-June

4.2 Domestic Manufacturing

The sector is overwhelmingly driven by Micro, Small, and Medium Enterprises (MSMEs), which account for nearly 90% of production. Uttar Pradesh, Punjab, Maharashtra, Delhi, Tamil Nadu, Jammu, and West Bengal are among India's top sports goods-producing states. Among these, legacy manufacturing hubs located primarily in Jalandhar (Punjab) and Meerut (Uttar Pradesh) produce 75% to 80% of the country's domestic output.

Figure 16: Commodity-Wise Share in India's Total Sports Goods Exports



Source: India Brand Equity Foundation. (2025, December). The sports industry and export in India.

<https://www.ibef.org/exports/sports-industry-india>

4.2.1 Profiles and Financials of Leading Domestic Manufacturers

Top domestic manufacturers such as Anand Track and Field Equipment (ATE), Bhalla International, and Nelco supplied official implements for the Tokyo Olympic Games. Furthermore, established brands

like Cosco, Nivia, Sareen Sports, and Sanspareils Greenlands (SG) continue to dominate domestic markets while exporting specialised gear worldwide.

Table 10: Overview of Leading Indian Sports Equipment Manufacturers

Manufacturer	Headquarters	Specialization	Revenue
Sanspareils Greenlands (SG)	Meerut, UP	This manufacturer specialises in premium cricket gear and leather test balls.	₹356 Crore
Nivia Sports	Jalandhar, Punjab	This brand specialises in inflatable footballs, basketballs, and athletic footwear.	₹372 Crore
Cosco (India) Limited	New Delhi	This publicly traded company specialises in tennis balls, court balls, and commercial fitness infrastructure.	₹174 Crore
Bhalla International (Vinex)	Meerut, UP	This enterprise specialises in certified professional track and field athletic equipment.	₹13.6 Crore
Sareen Sports Industries (SS)	Meerut, UP	This premium heritage brand specialises in custom English and Kashmir willow cricket bats for professional global athletes.	NA

Source: Corporate financial disclosures, Ministry of Corporate Affairs (MCA) filings, and company annual reports

India has set monumental targets for the next decade, closely aligned with its ambitions to host the 2036 Olympic Games. According to the NITI Aayog report, the Indian government aims to exponentially scale the country's sports equipment exports to an ambitious \$8.1 billion by 2036, an expansion projected to generate roughly 5.4 million jobs. This is complemented by the Draft National Sports Policy 2024, which is centred on the vision of 'Sports for Nation Building,' aiming to leverage sports for broad-based economic development, social inclusivity, and integration with the National Education Policy (NEP) 2020 to impart physical literacy from early childhood.

4.2.2 Structural Bottlenecks

Modern sports gear relies heavily on advanced materials like carbon fibre composites, polyurethane (PU), and EVA foam, which are not produced at scale domestically. Advanced composite elements like

carbon fibre fabrics, tubes, and raw sheets, the critical foundational material for modern tennis rackets, hockey sticks, and high-end athletic implements, are classified explicitly under HS Code 68151300 ("Other articles of carbon fibres"). To transition from a contract manufacturer to a global leader, India must overcome significant structural and economic hurdles that place it at a disadvantage compared to regional competitors. Importers face an aggressively compounded tariff structure. Rather than a basic flat tariff, the Central Board of Indirect Taxes and Customs (CBIC) applies an escalating stack of assessments that pushes the true cost of import significantly higher than the baseline rate (Table). Furthermore, India currently lacks adequate internationally accredited testing laboratories. Consequently, manufacturers must send products to Europe for certification, which can cost between ₹5 lakh and ₹50 lakh per specific model, making it largely unaffordable for MSMEs. The branding gap also persists, as India has very few globally recognised sports brands outside of cricket, with most firms acting as contract manufacturers for foreign companies.

Table 11: Effective Import Duty Stack for Carbon Fibre Articles (HS 68151300)⁵

Duty Component	Statutory Rate	Effective Cost (per ₹100 of CIF Value)	Calculation Formula
Basic Customs Duty (BCD)	10.00%	₹10.00	AV * 10%
Agriculture Infrastructure & Dev. Cess (AIDC)	10.00%	₹1.00	BCD * 10%
Social Welfare Surcharge (SWS)	10.00%	₹1.10	(BCD + AIDC) * 10%
Integrated GST (IGST)	18.00%	₹20.18	(AV + BCD + AIDC + SWS) * 18%
Total Compounded Effective Duty	—	32.28%	Sum of all component duties

Source EximPe. (2026). HSN Code 68151300 – Other articles of carbon fibres. HSN Code Finder. Estratto il 18 giugno 2026 da <https://eximpe.com/hsncode-finder/68151300>

While the statutory Basic Customs Duty (BCD) on advanced raw materials ranges between 10% and 20%, the true operational bottleneck for Indian MSMEs lies in the compounded tax stack. When factoring in infrastructure cesses and the standard 18% IGST, the total effective duty an importer must pay at the port scales to 32.28% (as shown in Table 8).

4.2.3 Strategic Interventions and Global Investments

Recognising these challenges, strategic initiatives are being mobilised across the public and private sectors to enhance India's global sports aspirations. The NITI Aayog report recommends rationalising import duties on critical raw materials and allocating ₹500 crore to a 'Brand India' initiative to improve international marketing visibility. The Sports Goods Export Promotion Council (SGEPC) actively provides financial schemes like the Market Access Initiative (MAI) and Market Development Assistance (MDA) to subsidise global marketing and product certification for MSMEs. To solve

⁵ Note: \$AV\$ = Assessable Value / Cost, Insurance, and Freight (CIF) value.

logistical inefficiencies, there are plans to develop new port-proximate greenfield manufacturing clusters in states like Gujarat, Andhra Pradesh, and Tamil Nadu. The future of sports equipment is highly technological, and institutions like IIT Madras are stepping up to incubate at least 200 deep-tech sports start-ups over the next five years. Finally, global retail and sportswear giants are recognising India's potential; Decathlon has planned a \$111 million investment to expand its local manufacturing. Decathlon's local footprint is massive: it plans 90 factories by 2030 to reach 90% "Made in India" for its Indian customers. Taiwan's Pou Chen Group has pledged an investment of ₹2,302 crores to establish a manufacturing facility in Tamil Nadu, creating 20,000 job opportunities over the next 12 years. Through these concerted efforts, India is actively positioning itself to command a significant share of the global sports economy.

Table 12: India's Carbon Fibre Industry Outlook (2025-30)

Metric	Value	Source
Market Size (2030)	\$908.5 million	Industry ARC
CAGR (2024–2030)	8.30%	Industry ARC
Major Domestic Producers	Reliance Industries, Jindal Advanced Materials, Silcarb Recrystallised	Multiple sources
Government Missions	National Technical Textiles Mission, Automotive Mission Plan 2047	MoT / MoHI
Major Demand Sectors	Wind Energy, EVs, Aerospace, Defence, Technical Textiles	

Chapter 5: Carbon Fibre in Indian Sport

5.1 Current Usage Across Indian Sports Disciplines

The current utilization of carbon fibre and advanced composites across Indian sports disciplines is characterized by a stark divide between elite professional access and grassroots affordability.

5.1.1 Racket Sports

Despite these supply chain challenges; racket sports represent one of the most active sectors for domestic carbon fibre integration in India. Pioneering domestic manufacturers, such as Arvind Composites, have successfully localized the production of high-performance carbon and glass fibre reinforced sports goods. Utilizing imported raw materials from Japan, South Korea, and Taiwan, these Indian facilities are producing 100% graphite badminton racquets (utilizing 24T to 40T Graphite) that weigh as little as 72 grams. By leveraging robotic and CNC equipment, domestic producers are now supplying customized, ultra-lightweight tennis and badminton rackets that cater to everyone from beginners to professional athletes, ensuring precise balance, variable flexibility, and high-tension capabilities.

5.1.2 Field Hockey

In field hockey, a sport deeply intertwined with India's sporting legacy, the transition from traditional wooden sticks to advanced composites is accelerating rapidly. To overcome the high costs of pure carbon fibre while still delivering elite performance, Indian composite manufacturers are developing advanced hybrid stick constructions. Domestic production lines are currently manufacturing hockey sticks that strategically blend 24T Carbon with high-strength E-Glass. This allows local brands to offer highly specialized equipment featuring various 'bow' profiles (such as Extreme Late Bows for powerful drag-flicking) and specific head shapes (like Maxi or Scoop bends) that provide Indian players with the necessary pop, consistency, and 3D skill control required in modern turf hockey.

5.1.3 Cricket

In cricket, where the International Cricket Council (ICC) strictly mandates the use of traditional wooden blades to preserve the spirit of the game, the application of carbon fibre and advanced polymers is ingeniously redirected towards protective equipment and handle reinforcements. Legacy manufacturing clusters in Meerut and Jalandhar, which produce the vast majority of India's sporting goods, are actively integrating advanced composites and high-density foams into the outer shells of batting helmets, as well as the inserts of wicket-keeping pads, thigh guards, and gloves. Leading domestic brands such as Cosco, Nivia, and Sanspareils Greenlands (SG) utilize these advanced materials to provide batsmen with

crucial vibration dampening and severe impact resistance against leather balls travelling at extreme speeds, achieving maximum safety without adding restrictive weight,

5.1.4 Track & Field

In track and field disciplines, Indian equipment manufacturers are steadily gaining global recognition and breaking into elite competitive circles. Notably, domestic legacy brands such as Anand Track and Field Equipment (ATE), Bhalla International, and Nelco have achieved world-class standards, supplying 36 official throwing implements (shot put, discus, and hammer) for the Tokyo Olympic Games. While athletes like Neeraj Chopra utilize foreign composite brands like Nordic or Nemeth for their specific aerodynamic properties, fellow Indian Olympians such as Tajinderpal Singh Toor and Seema Punia have competed using high-quality implements manufactured domestically in Meerut and New Delhi. This highlights a growing momentum where Indian manufacturing is evolving from basic traditional goods to globally certified, high-performance athletic equipment.

5.1.5 The Paralympic Opportunity

India's Paralympic movement, which delivered its best-ever medal haul of 19 medals at the Tokyo 2020 Games and followed it with 29 medals at Paris 2024, represents one of the most underleveraged opportunities in the domestic carbon fibre sports equipment landscape. Despite this remarkable competitive surge, Indian Paralympic athletes remain almost entirely dependent on imported carbon fibre running blades⁶, racing wheelchairs, and adaptive sports prosthetics primarily sourced from global manufacturers such as Össur, Otto Bock, and Ottobock at costs ranging from ₹3 lakh to ₹25 lakh per unit. Domestically, no MSME cluster in Jalandhar, Meerut, or elsewhere currently manufactures certified Paralympic-grade carbon fibre blades or adaptive composite equipment at scale, representing a critical gap that is simultaneously a social equity failure and a missed industrial opportunity. Given that carbon fibre prosthetic blade manufacturing shares substantial process overlaps with hockey stick and racket composite fabrication, both relying on multi-directional carbon layups, autoclave curing, and precision CNC finishing, existing domestic composite manufacturers like Arvind Composites are structurally well-positioned to diversify into this segment with relatively modest retooling investment. A targeted government intervention, potentially through the Sports Authority of India (SAI) and the Paralympic Committee of India (PCI) in conjunction with the National Technical Textiles Mission, to co-fund pilot production of domestically manufactured running blades and racing wheelchairs with strong export potential to the broader South and Southeast Asian adaptive sports market.

⁶ Light, springy and compact, the prosthetics, or running blades, are used by amputee athletes to replace the calf and ankle. The J-shaped prosthetic is made of carbon fibre, a strong, lightweight material, and helps an athlete as they step forward, whether during a sprint, a vertical lift or a long jump.

5.2 Import Dependency and Equipment Market Size

5.2.1 Domestic Sports Equipment Market

India's domestic sports goods market currently stands at an estimated \$2.5 billion, with the equipment segment comprising nearly \$0.5 billion of that total. Despite a profound socio-cultural connection to sports, India's footprint in the global sports equipment trade remains exceptionally small; the nation currently exports between \$275 million and \$523.24 million worth of equipment annually, capturing a mere 0.5% of the global export market. However, the growth trajectory is highly optimistic. Industry analysts project India's overall sports goods market to expand to \$6.6 billion by 2027, driven by a combination of surging domestic demand, health awareness, and aggressive government targets aiming to scale sports equipment exports to an ambitious \$8.1 billion by the 2036 Olympics.

5.2.2 Market Sizing for Carbon Fibre Niches: Badminton and Cycling

Within this broader market, segments that heavily utilize advanced composites like carbon fibre are experiencing explosive, high-value growth. For instance, the global carbon fibre badminton racquet market alone was valued at \$1.8 billion in 2025 and is projected to surge to \$3.2 billion by 2034, registering a compound annual growth rate (CAGR) of 6.6%. The Asia Pacific region completely dominates this niche, accounting for nearly 48.7% of total global revenue, heavily fuelled by mass-market adoption in countries like China, Indonesia, and a rapidly accelerating consumer base in India. The Carbon Fibre Bike Market is expected to grow from USD 1.4 billion in 2023 to USD 3.5 Billion by 2031, at a CAGR of 12% during the forecast period⁷. These figures underscore that advanced material integration is no longer a luxury segment, but a primary driver of overall sports market valuation.

5.2.3 Import Dependency on Indian MSMEs

Despite the massive economic potential of advanced sports equipment, India's manufacturing sector which is almost 90% driven by Micro, Small, and Medium Enterprises (MSMEs) based in hubs like Jalandhar and Meerut is critically handicapped by a total reliance on imported raw materials. Modern high-performance sports gear requires specialized polymers, performance fabrics, EVA foam, polyurethane, and, most crucially, aerospace-grade carbon fibre composites. Currently, India does not produce commercial raw carbon fibre domestically, forcing manufacturers to rely 100% on imports from a concentrated group of supplier nations, including the United States, France, Japan, and Germany. This absolute dependency exposes domestic manufacturers to severe geopolitical supply chain risks, currency fluctuations, and intense cost pressures.

⁷ Report Prime. (2026, April). *Carbon fiber bike market size, growth, forecast till 2032* (Report No. r17767). <https://www.reportprime.com/carbon-fiber-bike-r17767>

5.2.4 The Cost Disadvantage and the Push for Indigenisation

While domestic Jalandhar-cluster manufacturers can build high-carbon components (such as the 95% carbon SNS Pro Tour) at highly competitive baseline price points, they remain structurally constrained by an effective 32.28% import duty stack on raw carbon fibre inputs (HS Code 68151300). Because India completely lacks a domestic upstream production pathway for PAN-based precursors, our domestic manufacturers are trapped in a low-margin tier, facing a 6x to 14x retail price gap against elite imports that control the highly lucrative professional segments of the market.

5.3 The Elite/Grassroots Divide

5.3.1 The Paradox

India's recent sporting successes at the global level highlight a stark paradox within its equipment landscape: a severe divide between elite access and grassroots affordability. While top-tier athletes like javelin thrower Neeraj Chopra and badminton star Lakshya Sen have brought India unprecedented global sporting recognition, they frequently must rely on highly engineered, imported composite equipment to compete at the highest levels. For instance, elite Indian javelin throwers often utilize premium foreign brands such as Nemeth or Nordic to capitalize on their specific aerodynamic properties and advanced material construction. Conversely, grassroots athletes, youth academies, and developing players across the nation are often forced to train with heavier, traditional, or inferior materials. This lack of access to advanced composites puts emerging domestic talent at a distinct disadvantage from the very beginning of their athletic careers.

5.3.2 Bridging the Gap

Despite these severe hurdles, there are active, concerted efforts to democratise access to advanced sports equipment across India. To bridge the affordability gap without entirely sacrificing performance, domestic manufacturers are increasingly turning to advanced hybrid composites. For example, pioneering firms like Arvind Composites are producing badminton and tennis rackets utilizing a strategic blend of 24T to 40T graphite and glass fibres.⁸ This allows them to offer lightweight equipment ranging from 220-gram entry-level rackets to stiffer 370-gram professional frames to amateur and intermediate players at accessible price points.

Furthermore, the quality gap between domestic manufacturing and elite performance is slowly closing. During the Tokyo Olympic Games, three Indian manufacturers, Anand Track and Field Equipment (ATE), Bhalla International, and Nelco achieved world-class standards, supplying 36 official throwing

⁸ Arvind Composites is a **FRP Products Manufacturer** and part of Arvind a \$2 billion group with interests in Textiles, Retail, Engineering, Real Estate and Environmental Solutions. Arvind Composites started its operation in 2012, when [Arvind](#) established a glass fabric weaving facility one of its first foray into Technical Textiles. Pultrusion, Hand Moulding, Moulded Gratings, Carbon pre-preg Moulding and Continuous Lamination capabilities were added subsequently.

implements (shot put, discus, and hammer) for the event. Notably, elite Indian Olympians such as Tajinderpal Singh Toor, Kamalpreet Kaur, and Seema Punia utilized these high-quality, domestically produced implements on the Olympic stage rather than opting for foreign brands. Coupled with the Union Government's *Draft National Sports Policy 2024*, which explicitly emphasises establishing equitable access to sports infrastructure and comprehensive talent development for underrepresented and grassroots populations, India is laying the necessary groundwork to ensure that high-performance equipment will no longer be exclusively reserved for the elite.

5.4 Athlete and Coach Perspectives

5.4.1 The Drive for Competitive Advantage

Athletes are acutely aware of the performance advantages provided by carbon fibre composites, sometimes going to extreme lengths to access them. In the elite marathon running community, the performance benefits of carbon-plated 'super shoes' are perceived to be so significant that there is anecdotal evidence of athletes breaking their sponsorship contracts, or sponsors allowing their athletes to wear disguised competitor brands, simply to avoid being held at a competitive disadvantage. For elite field athletes, there is a strong preference for arriving at global competitions with highly customised, premium foreign composite equipment; for example, Indian javelin thrower Neeraj Chopra at the last moment changed his javelin - swapping the Nemeth with one known as the Nordic⁹ (Fig) to capitalise on their specific aerodynamic engineering rather than using standardised stadium equipment. Paralympic athletes share this perspective on the absolute necessity of advanced materials. Richard Whitehead MBE noted that carbon running blades provide the essential 'platform for success' that all athletes desire, while humbly acknowledging that the athlete's own dedicated training and effort remain the primary reasons for actual improvement in running times.

⁹ The two javelins differ in the nature of their composition -- the Nordic has a higher proportion of carbon fibre -- and in the shape of the landing tip -- the Nemeth being more rounded.

Figure 17 New Valhalla Composite javelins by Nordic Sport as used by Neeraj Chopra



Source: Selvaraj, J. (2021, March 5). Neeraj Chopra's Nordic weapon: Breaker of storms, and national records. ESPN. https://www.espn.in/athletics/story/_/id/31010150/neeraj-chopra-nordic-weapon-breaker-storms-national-records

5.4.2 The Custom Composite Shift in Indian Field Hockey

Underpinning the athlete and coach perspectives on material performance is a clear shift away from traditional wood toward advanced composites at the elite level. Following the Indian hockey teams' historic performances at the Tokyo Olympics, local manufacturers in Jalandhar's sports cluster, specifically *Rakshak Sports and Pooja Enterprises (Alfa)*, revealed that senior players like Manpreet Singh, Rani Rampal, and PR Sreejesh now bypass standard wooden sticks entirely¹⁰. Instead, demand ultra-light, custom-cured composite sticks precision-tailored down to the exact gram, typically ranging between a mere 535 and 550 grams. From the athlete's perspective, configuring the stiffness and structural mould of a carbon-composite matrix is vital for generating maximum kinetic strike velocity and maintaining rapid 3D ball handling on the pitch. Concurrently, the teams' Olympic success using these domestic designs fully validated the technology in India. This immediately triggered a 30% boom in market demand, prompting regional coaches and training academies to rapidly transition junior players toward modern composite gear to match the high-velocity tactical demands of the international game.

5.4.3 Subjective Feel vs. Objective Performance

Interestingly, an athlete's subjective perception of a carbon fibre product does not always align with objective performance data. In a study comparing running economy across seven different carbon-plated racing shoes, researchers found that athletes were generally poor at identifying which shoe actually made them the most efficient. Only 2 out of 12 subjects correctly chose the shoe that was most metabolically economical for them for a 5km race, often subjectively gravitating toward the lightest shoe rather than heavier, but more efficient, models. Conversely, in table tennis, player preference

¹⁰ <https://www.newindianexpress.com/india/2021/Aug/10/made-in-punjab-hockey-sticks-behind-indias-tokyo-magic-2342852.html>

strongly aligns with objective data and individual playstyle. A study evaluating elite table tennis players found that 70% preferred carbon rackets over traditional wooden ones, a choice that directly correlated with the statistically significant increases in ball speed and spin rates the carbon blades provided. Because carbon rackets were also shown to reduce control accuracy, coaches emphasize the need to carefully assess a player's skill level, recommending carbon materials strictly for advanced players who favour aggressive, offensive gameplay.

Chapter 6: India's Composite Manufacturing Opportunity

6.1 Current State of Composite Manufacturing in India

6.1.1 Market Size and Material Dominance

The Indian composite materials market is on a robust growth trajectory, but it remains structurally nascent compared to mature global markets. In 2021, the domestic market was valued at \$1.26 billion, representing a consumption of 511.9 kilotons, and is projected to reach \$2.01 billion (768.2 kilotons) by 2027. Despite this steady expansion, composite consumption in India is exceptionally low at just 0.37 kg per capita, presenting a stark contrast to highly penetrated markets like the United States (11.5 kg) and Germany (7.7 kg). Currently, the ecosystem is overwhelmingly dominated by glass fibre, which holds an unassailable 99.1% share of the reinforcement market on a volume basis. This dominance is largely driven by cost-sensitive industries such as ground transportation, building, and renewable energy. Similarly, polyester is the preferred matrix system, accounting for 70.8% of the market volume, while epoxy trails at 10%. In contrast, carbon fibre utilisation remains marginal, predominantly restricted to the aerospace and defence sectors or specific export-oriented automotive and sporting applications with almost all carbon fibre imported in the form of prepregs.

6.1.2 Cost of Technology

A critical bottleneck crippling India's advanced composite ecosystem is the complete absence of indigenous raw carbon fibre production. The country currently relies 100% on imports from nations such as the United States, Japan, France, and Germany. The cycle begins with reliance on global suppliers across Japan, the United States, and Europe to secure primary raw materials, including polyacrylonitrile (PAN) precursor and specialized carbon fibre tow. These materials must travel through complex global shipping lanes and maritime logistics networks before reaching Indian ports, leaving domestic production timelines highly exposed to international supply chain shocks, container shortages, and sudden freight tariff spikes. Historically, attempts to indigenize high-performance carbon fibre have proven to be technically complex and capital-intensive. For example, a long-term research initiative was led by the National Aerospace Laboratories (NAL) and the Indian Space Research Organisation (ISRO) to develop aerospace-grade T800 carbon fibre as an import substitute, but due to delays in procurement of equipment and other technical problems that occurred during the execution, the project had to be extended five times up to 2015, which was indicative of improper planning. Further, despite repeated extensions, NAL could not develop the special grade (T 800 grade) Carbon fibres due to its

inability to support the project in terms of manpower¹¹. For India's Micro, Small, and Medium Enterprises (MSMEs), this import reliance has severe economic consequences.

6.1.3 Prevailing Fabrication Processes

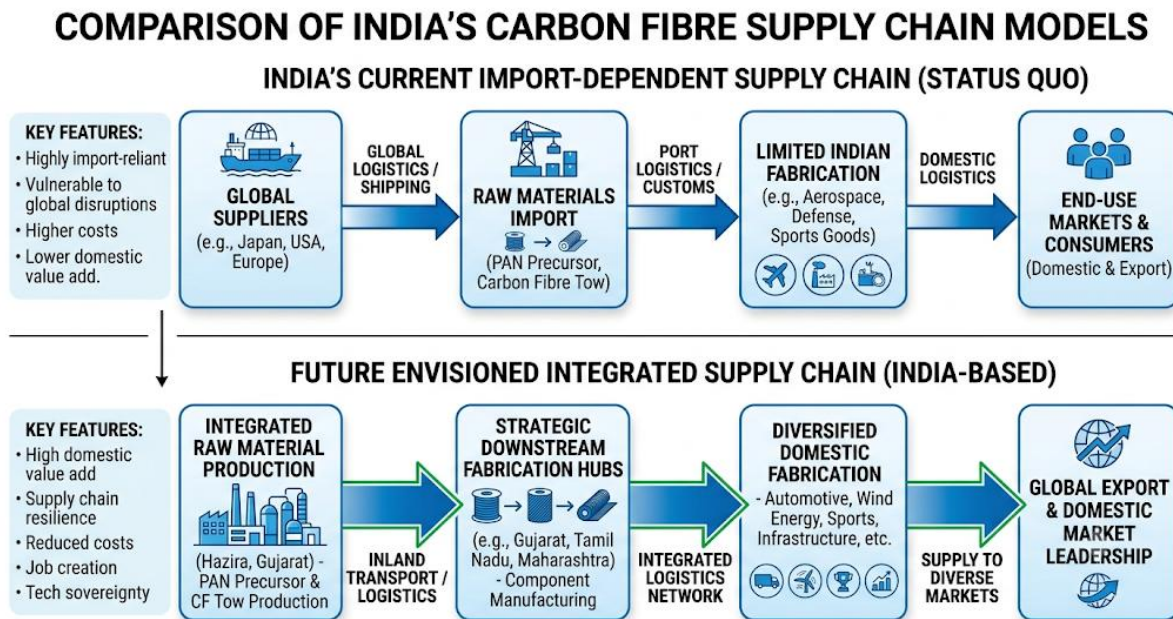
On the fabrication front, India's composite manufacturing landscape remains heavily reliant on traditional, labour-intensive methods. Open mould processes, such as manual hand and spray layups, dominate the industry, accounting for 41% of the total manufacturing process share in 2021. This high percentage reflects an industry heavily populated by small-sized fabricators that lack the capital required to invest in highly automated systems. Advanced prepreg layup processes are the standard for elite carbon fibre sporting goods and aerospace components, but they account for a mere 0.8% of the Indian market. However, there is a gradual modernisation occurring across the sector. Advanced, closed-mould processes are slowly gaining traction, with injection moulding (15.5%), compression moulding (13.7%), and resin infusion (11%) increasingly utilised for ground transportation components and wind turbine blade construction.

6.1.4 Strategic Shifts and Indigenisation Efforts

To overcome these systemic hurdles and bridge the global technology gap, massive strategic shifts are underway across both the public and private sectors. The Indian government has made indigenisation a priority, with Union Textiles Minister Giriraj Singh projecting that domestic commercial production of niche carbon fibre will successfully begin by 2025–26. This is supported by the National Technical Textiles Mission, which has sanctioned over 156 research projects dedicated to the development of carbon fibres and advanced composites. Pioneering domestic sporting goods manufacturers like Arvind Composites are demonstrating that India can achieve advanced fabrication locally; they are actively utilising CNC equipment, robotic systems, and in-house prepregs to manufacture world-class carbon and glass fibre rackets and hockey sticks. Together, these raw materials and fabrication advancements are poised to drastically reduce costs, eliminate import dependencies, and position India as a highly competitive global hub for advanced composite manufacturing.

¹¹ Public Accounts Committee. (2025). *Short closure of project for development of special grade carbon fibre (Department of Space)* (21st Report, 18th Lok Sabha). Lok Sabha Secretariat. http://web.archive.org/web/20250807085040/https://sansad.in/getFile/lsscommittee/Public%20Accounts/18_Public_Accounts_21.pdf?source=loksabhadocs

Figure 18: Comparative Analysis of Current (Import-Dependent) vs. Future Envisioned (Integrated) Carbon Fibre Supply Chains in India.



The diagram above illustrates the fundamental shift from India's traditional, import-reliant structure to an optimized, integrated, domestic supply chain model. This transition is essential for building long-term resilience and reducing localized production costs.

6.2 Comparison with China, South Korea, and Taiwan

6.2.1 Raw Material Independence vs. Import Dependency

The most critical divergence between India and its East Asian competitors, China, South Korea, and Taiwan, is in raw material sovereignty. China (Box 2), South Korea, and Taiwan have heavily invested in indigenous, large-scale production capabilities. While India's domestic manufacturing capability remains largely anchored in First and early Second-Generation processes, hand layup, manual fabrication, wood and aluminium hybrids, its East Asian competitors have fully industrialised Third Generation automated fibre placement, robotic layup, and smart composite integration. This framing makes the competitive gap feel structural and generational rather than merely financial. China's total carbon fibre production capacity reached an immense 135,500 metric tons in 2024, supported by aggressive capacity expansions from domestic manufacturers focusing on high-performance T700/T800-grade fibres. Similarly, South Korea (driven by major industrial players like Hyosung Advanced Materials) and Taiwan (led by Formosa Plastics Corporation) have established themselves as key global suppliers of raw carbon fibre (Box 6.2.1).

Table 13: Comparative Analysis of China, South Korea, and India's Adaptation Pathway

Parameter	China (Supply-Side)	South Korea (Demand-Side)	India (Current Position)
Core Strategy	State-backed domestic carbon fibre production; vertical integration from PAN precursor to finished sports goods	Demand creation through elite sport investment; high-performance equipment procurement tied to national teams	Import-dependent;
Primary Policy Instrument	14th Five-Year Plan classification of carbon fibre as a strategic material; regional manufacturing subsidies	National sports federation procurement agreements; OEM ¹² -supplier partnerships for composite equipment	National Technical Textiles Mission (NTTM); PLI scheme for technical textiles
Key Private Actor	Zhongfu Shenying, Jilin Carbon (upstream); Li-Ning, Victor (downstream sports brands)	Hyosung Corporation (carbon fibre); domestic sports brands supported through federation contracts	Reliance Industries (upstream, pre-production); Arvind Composites (downstream fabrication)
Sports Equipment Anchor	Li-Ning CBA supplier agreement (2024); Olympic medal pipeline as brand platform	National team equipment indigenisation targets, carbon fibre use in archery, shooting, and winter sports	No formal federation-manufacturer linkage; elite equipment is still 100% imported
Carbon Fibre Production Capacity	~30,000+ MTPA domestic capacity; near self-sufficient	Limited domestic precursor; strong downstream composite fabrication	0 MTPA currently; Reliance plant planned at 20,000 MTPA
Transferability to India	IIT research ecosystem mirrors China's industry-academia model	The Green Hydrogen Mission and sports federation structure provide equivalent demand anchors	Needs both upstream investment acceleration and downstream federation-industry linkage

¹² Original Equipment Manufacturer

Box 6.2.1: How Taiwan Built a Carbon Fibre Badminton Racket Industry

Taiwan built its dominant carbon fibre racket industry by securing raw material sovereignty through state-backed petrochemical investments, notably Formosa Plastics, which established its Tairyfil facility by 1984 with an eventual annual capacity of 8,750 metric tons. This upstream foundation eliminated import reliance, allowing domestic brands like Victor, which now records \$440 million in revenue and global giant Yonex, which set up its Taiwan hub in 1987, to capture a market projected to reach \$303.1 million by 2030.

6.2.2 Market Dominance in Sports Composites

In the specialised sports equipment market, particularly for advanced carbon fibre products, China, South Korea, and Taiwan are the undisputed dominant forces. In the lucrative badminton sector, China alone accounts for an estimated 30% of the Asia Pacific region's total carbon fibre racquet volume. This dominance is not limited to racket sports; when the global football industry transitioned to advanced

Box 6.2.2: China's Carbon Fibre Badminton Racket Industry

China's dominance in carbon fibre sports equipment was built on state-directed investment in domestic PAN-based carbon fibre production through the 1990s and 2000s, which by 2010 had made Chinese composite manufacturers largely independent of Japanese suppliers like Toray. From 1992 onwards, the Chinese badminton team won gold medals at the Olympic Games. China's unbroken badminton dominance from 1992 onwards across the Olympics, World Championships, Thomas Cup, Uber Cup, and Sudirman Cup served simultaneously as a global brand platform and a live performance laboratory for its domestic carbon fibre sports equipment industry.

thermo-bonded (stitchless) technology, China rapidly adapted and effectively maintained a global monopoly on thermo-bonded ball supplies from 2002 to 2010 (**Box 6.2.2**). Conversely, manufacturers in the Indian subcontinent lost significant market share during this period because they hesitated to invest in the new technology, relying instead on traditional hand-stitching. The lesson for India is threefold: first, indigenise the raw material (carbon fibre precursor) before attempting to compete at the finished goods level; second, formalise the link between national sporting federations and domestic equipment brands through official supplier agreements; and third, treat elite athlete performance as a public demonstration platform for domestic manufacturing capability rather than an occasion to import foreign equipment.

6.3 Alignment with Make in India and PLI Schemes

6.3.1 The 'Make in India' Vision and Import Substitution

The push to develop indigenous carbon fibre and advanced composites is fundamentally intertwined with the Government of India's 'Make in India' and 'Atmanirbhar Bharat' (Self-Reliant India) initiatives. Historically, the advanced sports materials sector has been highly vulnerable due to a heavy reliance on imports, facing steep customs duties of 10% to 20% on critical raw materials like carbon fibre, EVA, and polyurethane. By explicitly integrating sports equipment and composite manufacturing into national schemes, the government aims to drive innovation, shield domestic manufacturers from geopolitical supply chain risks, and build a self-sustaining industrial ecosystem. Incentives under *Make in India* and *Startup India* are also actively enhancing Foreign Direct Investment (FDI) within the sports manufacturing sector.

6.3.2 Production-Linked Incentive (PLI) Schemes and Technical Textiles

The Production-Linked Incentive (PLI) scheme serves as a primary financial catalyst, strategically designed to reward scale, build vertical integration, and make domestic production globally cost-competitive. The PLI framework has been aggressively expanded to support advanced materials through the PLI Scheme for MMF Fabric, MMF Apparel, and Technical Textiles. Union Textiles Minister Giriraj Singh has highlighted that PLI interventions have already successfully balanced import deficits in allied sectors like hygiene products, proving the model's effectiveness and creating excitement across the industry. By deploying these financial incentives alongside the National Technical Textiles Mission (NTTM) which has sanctioned 156 research projects, including the indigenous development of carbon fibres the government is providing the critical risk coverage needed for domestic firms to transition into high-tech manufacturing by 2025–26.

6.3.3 Union Budget FY 2026-27 and Sector-Specific Support

The Union Budget FY 2026-27 heavily reinforced this momentum by introducing a dedicated 'Sports Goods Manufacturing Initiative,' designed to provide focused support to enhance manufacturing, research, and innovation specifically in sports equipment and material sciences. To address the 10% to 20% cost disadvantage Indian MSMEs face compared to regional competitors like China and Pakistan, NITI Aayog policy recommendations emphasize rationalising import duties on critical raw materials such as carbon fibre, polyurethane (PU), and EVA foams until domestic production capacity reaches maturity. Furthermore, recognizing that MSMEs form the backbone of this industry in traditional hubs like Jalandhar and Meerut, the government is focusing on modernising these existing clusters while developing new port-proximate manufacturing clusters in states like Gujarat, Andhra Pradesh, and Tamil Nadu.

6.3.4 'Brand India' and Ambitious Export Targets

These synergistic manufacturing policies are ultimately geared towards a highly ambitious economic target: scaling India's sports equipment exports to \$8.1 billion by the 2036 Olympics. To achieve this, NITI Aayog reports recommend allocating ₹500 crore to a 'Brand India' initiative aimed at improving the international visibility and reputation of domestically manufactured gear. Concurrently, the *Draft National Sports Policy 2024* emphasizes regulating the manufacturing and distribution of sports equipment to ensure that Indian products consistently meet high global certification standards, further enhancing market competitiveness. As a long-term strategic lever, policymakers are preparing the industry to meet the global standards and strict timelines required to supply equipment for major international sporting events, ensuring that the domestic manufacturing ecosystem receives guaranteed demand and an unparalleled global showcase.

Chapter 7: Emerging Trends Relevant to India

7.1 Barriers: Cost, Skilled Labour, and Technology Gaps

The transition to advanced composites is heavily impeded by extreme capital requirements and structural cost disadvantages. Standard-grade carbon fibre commands a premium price of approximately \$20 to \$25 per kilogram, making it exponentially more expensive than traditional aerospace-grade aluminium (\$2.5/kg) or low-carbon steel (\$1.1/kg). Furthermore, establishing a composite manufacturing facility requires massive capital investments in highly specialised equipment, such as autoclaves, precision CNC-machined moulds, and automated fibre placement systems, which cost millions of dollars.

For Indian manufacturers, these baseline costs are severely exacerbated by a 100% reliance on imported raw materials. Because India lacks commercial domestic carbon fibre production, Micro, Small, and Medium Enterprises (MSMEs) in hubs like Jalandhar and Meerut must import these advanced polymers, subjecting them to steep customs duties ranging from 10% to 20%. This heavy taxation drastically erodes the already thin profit margins of domestic producers, resulting in a fundamental cost disadvantage, whereas China benefits from immense economies of scale and an integrated domestic supply chain. Additionally, meeting international market standards requires certifications that can cost between ₹5 lakh and ₹50 lakh per Stock Keeping Unit (SKU), a financial hurdle that remains unaffordable for many MSMEs.

In traditional sporting goods clusters across the subcontinent, the industry remains deeply tethered to the conventional *ustaad-shagird* (master-helper) apprenticeship system. In this highly informal structure, young helpers are trained at a very slow pace, leading to a workforce with low formal literacy and virtually no technical institute qualifications. Manufacturers frequently prefer this cheaper, ad-hoc labour over qualified technicians, falsely perceiving that formally trained workers demand higher salaries and will waste valuable time and materials while adapting to specific factory floors. This reluctance to invest in permanent, formally educated personnel severely limits the sector's ability to scale up complex, precision-based composite production.

India's aspirations to manufacture high-end composites are significantly constrained by a lack of access to proprietary global technologies and fragmented domestic infrastructure. Historically, attempts by Indian state agencies to develop aerospace-grade (T800) carbon fibre have been thwarted by restrictive global 'technology denial regimes,' which block access to the critical, closely guarded equipment necessary to process ultra-high-strength polymers.

Furthermore, India severely lacks internationally accredited testing and certification laboratories, forcing local manufacturers to send their products to Europe for validation. The industry also suffers

from a broader lack of an integrated manufacturing ecosystem; production is heavily skewed towards downstream assembly rather than full-cycle fabrication, resulting in operational inefficiencies, logistical bottlenecks, and weak linkages with global supply chains. Finally, a historical hesitancy to adopt emerging technologies, such as thermo-bonded assembly or broad composite utilisation outside of field hockey, has repeatedly allowed agile competitors like China and Taiwan to capture lucrative global market shares.

7.2 Smart Composites and Embedded Sensing

The next frontier in sports equipment manufacturing is the transition from passive structural components to 'smart composites.' This involves the fusion of high-performance carbon fibre and advanced polymers with embedded electronics, micro-sensors, and interactive materials. By integrating performance analytics directly into carbon fibre gear, manufacturers are creating equipment capable of capturing real-time physiological and biomechanical data, transforming how athletes train, compete, and refine their techniques.

The integration of sensors into rigid carbon fibre equipment is already yielding tangible results across various sports. In the racket sports market, brands such as Babolat are developing sensor-integrated carbon fibre racquet handles recording stroke patterns. Similarly, modern carbon fibre racing shells are now equipped with highly sophisticated onboard computers and sensors that track each rower's performance. Coaches utilise gauges attached to the oarlocks to measure the exact amount of force each athlete exerts on their carbon fibre blade.

Beyond rigid equipment, the concept of embedded sensing is heavily influencing sportswear and adaptive athletic fabrics. Innovations in nanotechnology and smart materials are enabling the creation of 'interactive materials.' For example, the Commonwealth Scientific and Industrial Research Organisation (CSIRO) in Australia developed a wearable fabric sleeve equipped with embedded sensors that track a basketball player's movements (Image). When the player takes a shot, the sensors transmit real-time data to a computer, which responds with specific audio tones; this instantaneous feedback loop helps the athlete build correct muscle memory and identify poor movement patterns.

Figure 19: CSIRO's Dr Richard Helmer demonstrating the wearable body mapping sleeve



Source: CSIRO

Furthermore, research institutes like the Ahmedabad Textile Industry's Research Association (ATIRA) are actively developing graphene-based e-textile sensors. By harnessing graphene's distinctive electrical and structural qualities, these advanced textiles can monitor minute body deformations, paving the way for next-generation wearable technologies in sports monitoring, health, and safety. Experts anticipate the casual integration of these smart fabrics into Olympic sportswear to provide real-time physiological data and help athletes optimise their competition strategies.

The rise of smart sports equipment represents a highly lucrative, long-term growth opportunity for the carbon fibre market. Industry analysts recommend that composite manufacturers actively enhance their collaborations with technology firms and universities to co-develop these next-generation smart materials. While smart carbon fibre products currently represent a small fraction of overall market volume, their ability to provide credible, quantified performance narratives is expected to drive rapid adoption among tech-savvy consumers and advanced athletes, allowing brands to command premium pricing and expand their market appeal.

7.3 Additive Manufacturing and Custom Equipment

7.3.1 The Shift from Mass Production to Personalization

The sports equipment industry is undergoing a paradigm shift from mass-produced, one-size-fits-all manufacturing toward highly individualised gear, driven by the unique properties of advanced composites. Because carbon fibre is inherently anisotropic, its structural characteristics are not uniform; instead, they depend entirely on how the fibres are oriented. This allows manufacturers to strategically tailor the material's behaviour to match an individual athlete's biomechanics. For example, in racket sports, designers can meticulously adjust carbon fibre layups to create rackets with varying degrees of

stiffness across different areas of the frame, optimizing the balance of power, control, and vibration dampening for specific playing styles. Similarly, modern hockey stick manufacturers engineer do flex profiles, kick points, blade curves, and shaft finishes to perfectly complement how a specific player moves and shoots on the ice.

7.3.2 Custom Orthotics, Prosthetics, and Smart Integration

Carbon fibre is particularly transformative in the development of personalized sports prosthetics and orthotics. Leading sports and fitness goods manufacturers are increasingly integrating 3D printing technologies alongside Augmented Reality (AR) and Virtual Reality (VR) to create highly specialized equipment. These advanced digital tools allow engineers to map an athlete's physical dimensions and movement patterns in real-time,

7.3.3 Precision Engineering and Material Optimization

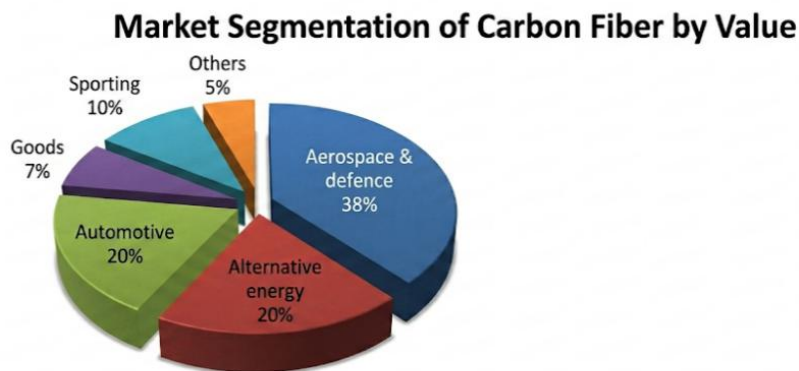
Brands are utilizing advanced computer modelling and automated manufacturing techniques to achieve hyper-customized performance. Technologies such as Precision Layer Design (PLD) utilized in elite hockey stick manufacturing employ highly calculated algorithms to dictate exactly where ultra-thin carbon sheets and resins should be placed or eliminated. This extreme precision ensures that custom equipment is not only perfectly tuned to the athlete but is manufactured with zero excess weight. As these additive and precision-layering technologies become more scalable, the athletes of tomorrow will benefit from gear that is fundamentally an extension of their own bodies, merging biomechanical data with advanced material science for unparalleled elite performance.

Chapter 8- Broader Industrial Applications of Carbon Fibre

While carbon fibre's revolutionary impact on elite sports equipment is profound, its true economic and technological value extends far beyond the playing field (Fig). The global manufacturing ecosystem is currently undergoing a massive transformation driven by megatrends such as sustainability, rapid decarbonization, and digitalisation. While traditional metals like steel and aluminium have historically dominated heavy industries due to their affordability, they are increasingly being replaced by "new-age" advanced materials. Carbon fibre has emerged as the premier substitute for these conventional load-bearing metals because it offers an unparalleled strength-to-weight ratio, exceptional stiffness, high-temperature tolerance, and superior corrosion resistance.

To fully understand the material's impact on the modern economy, it is essential to look beyond sporting goods. This chapter explores the expanding footprint and application versatility of Carbon Fibre-Reinforced Polymers (CFRP) across major global industries, detailing its critical integration in aerospace and defence, automotive and mass transportation, renewable energy, civil infrastructure, and advanced healthcare.

Figure 20: The main applications of carbon fibres globally: automotive, alternative energy, aerospace, transportation, building, sport and others.



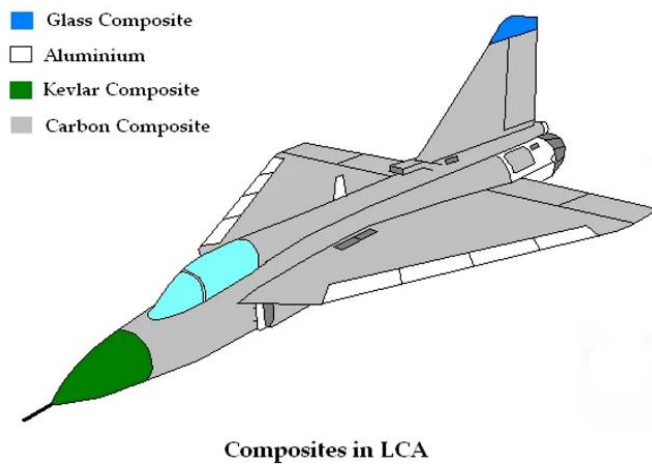
Source: Mordor Intelligence. (2026, January). Carbon fibre market size & share analysis - growth trends and forecast (2026 - 2031). <https://www.mordorintelligence.com/industry-reports/carbon-fibre-market>

8.1 Aerospace and Defence

In the commercial aviation sector, carbon fibre-reinforced polymers (CFRP) have emerged as the forefront material for modern aircraft construction. Driven by the industry's need for "lightweighting" and improved fuel efficiency, advanced composites are prominently used to manufacture aircraft fuselages, wings, tail parts, and interior structures. By replacing heavier traditional metals, carbon

composites offer an exceptional strength-to-weight ratio, excellent fatigue resistance, and superior dimensional stability across varying temperature environments.

Figure 21: The primary material in the Tejas is carbon-fibre reinforced composite



Source -WION. (2019, April 1). The HAL Tejas fighter jet is made of the same materials used to build spacecraft. WION News.
<https://www.wionews.com/photos/the-hal-tejas-fighter-jet-is-made-of-the-same-materials-used-to-build-spacecraft-1763794935889>

The defence sector heavily relies on carbon fibre to produce high-performance military assets. Carbon composites serve as a critical lightweight structural material in military fighter jets, notably utilized in India's Light Combat Aircraft (LCA) *Tejas* (Fig). Furthermore, carbon-carbon composites are frequently used to manufacture high-performance brake pads for light combat aircraft because their mechanical strength actually increases at higher temperatures, allowing them to withstand extreme thermal fatigue. The rapid rise of autonomous warfare has also driven

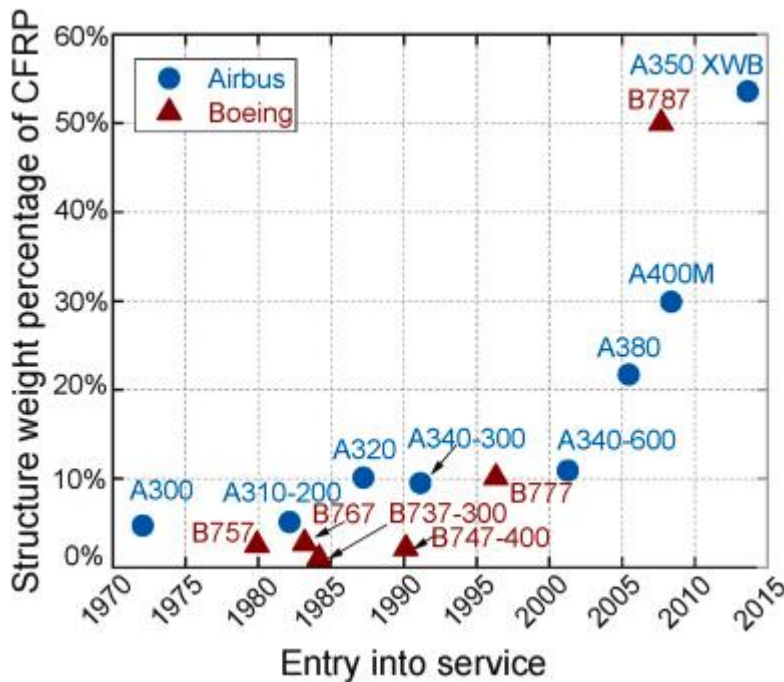
significant demand for carbon composites in the manufacturing of drones and unmanned aerial vehicles (UAVs), where lightweight structural integrity is absolutely critical.

For extreme environments, highly specialised grades of carbon fibre are indispensable. The Vikram Sarabhai Space Centre (VSSC) under the Department of Space relies on high-strength, aerospace-grade (T800) carbon fibre for critical launch vehicle programs, including upper stage motor cases, spacecraft pressuring tanks, and structural elements for the *Gaganyaan* mission. Additionally, specialised carbon-carbon composites and ceramic matrix composites (CMC) are deployed in India's Integrated Guided Missile Development Programme (IGMDP), such as the nose-tip of the *Agni* missile, and in spacecraft reentry vehicles. These materials are chosen because they offer tremendous thermal stability and retain their structural integrity at extreme temperatures without deformation.

Historically, India's aerospace and defence sectors have been highly vulnerable due to a strict reliance on importing strategic raw materials, like T800-grade carbon fibre, from foreign vendors on a proprietary basis. Procuring these advanced materials is often complicated by restrictive global "technology denial regimes". To counter these geopolitical risks and ensure national strategic security, there is a strong policy imperative to develop indigenous manufacturing capabilities. Domestic institutions, including the Defence Research and Development Organisation (DRDO) and the National

Aerospace Laboratories (NAL), are continuously working to develop special-grade carbon fibres and advanced composites to support India's self-reliant defence infrastructure.

Figure 22 Increased CFRP usage in commercial Airbus and Boeing aircraft.



Source: Engels, A. (2024). Development of a smart production line for large CFRP box structures (Doctoral dissertation, RMIT University).

8.2 Automotive and Transportation

8.2.1 The Need for Lightweighting and EV Integration

Driven by stringent global fuel economy and emissions regulations, the automotive industry has made "lightweighting" a central strategy. Replacing traditional cast iron and steel with carbon fibre-reinforced polymers (CFRP) can reduce a vehicle's weight by up to 50%, with every 10 kg reduction leading to a drop in carbon emissions of 1 g/km. Table provides comparison of weight of steel and CFRP. This weight reduction is particularly critical in the rapidly expanding Electric Vehicle (EV) market. Because EVs carry heavy lithium-ion battery packs, lightweighting is absolutely essential to offset battery mass and extend the vehicle's driving range, with a 10% overall weight reduction capable of yielding a 6–8% range increase. Leading this paradigm shift, vehicles like the BMW i3 have heavily integrated CFRP into their mass-produced bodies, achieving structural assemblies that are 50% lighter than steel and 30% lighter than aluminium.

Table 14: Comparison of weight reduction in the car roof by replacing steel with CFRP.

Description	Weight
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Steel roof	11.2 kg
Composite roof (CFRP)	5 to 5.5 kg
Weight reduction	6 to 6.6 kg

Source: Masilamani R, Dhandapani N, Kumar K, Mani K 2017 A review on usage of carbon fibre reinforced plastics in automobiles. International Journal of Pure and Applied Mathematics 117 (20) 537-44

8.2.2 Carbon Fibre Global Demand in Automotive Market (2024)

Metric / Feature	North America	Europe	Asia-Pacific	Middle East & Africa
Global Market Share	21%	27%	48% (Dominant)	4%
Primary Key Drivers	EV production, lightweight trucks, and advanced composite integration.	Premium manufacturing, sustainability regulations, and e-mobility adoption.	Massive EV manufacturing expansion, lightweight vehicle production.	Assembly ops growth, e-mobility investments, lightweight transit projects.
Key Regional Hubs	USA (Largest market); (46% of regional production).	Germany, Italy, France, UK (Major markets).	China (41% of global carbon fibre production capacity), Japan, South Korea, and India.	Gulf Countries (43% of regional composite utilisation).
Passenger Vehicle Demand	71% of domestic demand	76% of European demand	75% of regional demand	68% of regional demand
Commercial Vehicle Share / Applications	28% of regional demand (lightweight truck & logistics chassis).	EV battery enclosure applications (thermal-resistant structures).	Commercial electric bus production (public transit electrification).	Premium assembly activities and government-supported transit projects.

Growth in Recyclable CF Tech	Up 19%	Up 24% (driven by stricter regulations)	Up 27% (driven by environmental targets)	Expanding (backed by sustainability-focused transit projects).
Automation & Tech Efficiency Gains	+15% production efficiency via robotic assembly systems.	+16% production accuracy via automated robotic assembly.	+14% thermal resistance via advanced resin systems (for battery/chassis).	+13% production precision via automated composite assembly.

360 Research Reports. (2024). *Carbon Fibre in Automotive Market - Global Market Insights and Forecast to 2030 (Report ID: 216099)*. <https://www.360researchreports.com/market-reports/carbon-fibre-in-automotive-market-216099>

8.3 Construction and Civil Infrastructure

8.3.1 The Transition to Advanced Composite Concrete

While traditional construction relies heavily on standard steel and concrete, these materials are increasingly limited by their susceptibility to environmental degradation and immense weight. Carbon fibre is disrupting this sector by offering tensile strengths ranging from 3,500 to 7,000 MPa, drastically outperforming standard structural steel, which typically offers only 400 to 550 MPa. To leverage this strength, the industry is increasingly adopting Carbon Fibre-Reinforced Concrete (CFRC) and Carbon Fibre-Reinforced Polymers (CFRP). By replacing heavy, rust-prone steel rebars with carbon fibre composites, engineers can construct highly durable structures that completely eliminate the risks of oxidation and chloride-induced corrosion.

8.3.2 High-Performance Carbon Fibre Structural Reinforcement Systems

According to a study by Alsuhaibani (2024), builders can find the exact right layout for Carbon Fibre Reinforced Polymer (CFRP) sheets to get the best support out of concrete beams while saving on materials. Testing shows that adding carbon fibre sheets drastically strengthens concrete beams, especially when using specific placement setups. For example, using a full-length sheet secured at the ends with a double U-shaped wrap gives the beam an incredible mechanical boost, achieving the highest weight capacity, a 161% improvement and increasing energy absorption by 822%. This design consistently beats other layouts across all key measurements, allowing the beam to bend 192% further before breaking while only slightly changing its initial stiffness. This proves why planning the exact dimensions of the carbon fibre sheet is vital to unlocking the best overall performance.

Figure 23: Application of CFRP in strengthening RC Beams



Source: Horse Construction. CFRP Sheets for Flexural Strengthening of RC Beams.
<https://www.horseen.com/solution/CFRP-sheets-for-flexural-strengthening-of-RC-beams>

8.4 Renewable Energy and Wind Turbines

8.4.1 Scaling Up Wind Energy Infrastructure

As the global economy accelerates its transition toward clean power, the renewable energy sector particularly wind power has become one of the largest and fastest-growing consumers of carbon fibre. To meet ambitious climate targets, such as India's mandate to install 500 GW of renewable capacity by 2030, the wind industry is designing ever-larger turbines to maximise energy capture. Modern next-generation turbines (in the 11–22 MW class) feature colossal rotor blades ranging from 100 to over 140 metres in length. At these extreme dimensions, traditional fibreglass becomes restrictively heavy and flexible, placing immense mechanical stress on the turbine's tower and drivetrain.

8.4.2 The Role of Carbon Fibre Spar Caps

To overcome these structural limitations, manufacturers are strategically integrating carbon fibre into the main load-bearing components of the blades, known as "spar caps". Because carbon fibre offers a significantly higher specific modulus (stiffness) than E-glass, it enables the construction of longer, slenderer, and highly aerodynamic blade designs. Studies have demonstrated that replacing fibreglass with carbon fibre spar caps can reduce the overall mass of a wind turbine blade by 20% to 25%. This substantial weight reduction not only lowers tower-top loads and installation costs but also enhances the overall energy output and operational longevity of the turbine. The table below provides a strategic comparison of three distinct materials (Standard Fibreglass, Standard Carbon Fibre, and Novel Heavy Tow Textile Carbon Fibre) evaluated for wind turbine blade spar caps, the critical internal spine supporting the blade against bending forces.

Table 15 Comparative Performance and Cost Metrics for Wind Turbine Spar Cap Materials

Material	Weight	Material Cost	Overall Verdict
Fibreglass	Heavy	Baseline	Reliable, but heavy and currently more expensive than the new option.
Standard Carbon Fibre	Light	48% Higher than Fibreglass	Too expensive to justify for most designs.
New Heavy Tow Carbon	Light	16% Lower than Fibreglass	New Heavy Tow Carbon saves 16% on a structural/system-level cost basis

Sandia National Laboratories. (2019). *Optimized Structural Designs for Low Wind Speed Turbine Blades* (Report No. SAND2019-14173), p. 30. U.S. Department of Energy. Retrieved from <https://www.energy.gov/sites/prod/files/2019/12/f69/SAND2019-14173-Optimized.pdf#page=30>

8.4.3 Offshore Durability and Cost Innovations

Carbon fibre's advantages are especially critical in the rapidly expanding offshore wind sector. Offshore turbines must endure highly corrosive marine environments and extreme weather conditions; in typhoon-prone regions, carbon composites have been shown to withstand cyclic fatigue 30% to 40% better than traditional glass fibres. While the high cost of carbon fibre has historically been a barrier for wind manufacturers, new innovations such as the development of textile-based carbon fibres that cost 40% less than commercial grades are making the material more economically viable. Furthermore, domestic manufacturing is scaling up to meet this demand. For instance, in October 2025, Kinenco Exel Composites India (KECI) commenced full-scale production of internationally certified carbon fibre wind components at its Banda facility, directly supporting the localized expansion of India's renewable energy infrastructure.

8.5 Healthcare and Biomedical

Carbon fibre is highly valued in medicine for its radiolucency, allowing X-rays to pass through without interference or scattering. This makes it ideal for manufacturing imaging machine beds, scanning supports, and oncology implants (like CFR-PEEK), which drastically reduce imaging artefacts on post-operative CT and MRI scans compared to traditional metals. Furthermore, CFR-PEEK composites are emerging as superior alternatives to titanium or stainless-steel orthopaedic implants. They closely mimic the elasticity of human bone, which minimises stress shielding and accelerates fracture healing times while offering excellent biocompatibility that reduces the risk of implant rejection. Beyond rigid implants, carbon fibre mesh serves as a three-dimensional scaffold in tissue engineering to promote cell growth for tissue repair. In India, institutions like the Ahmedabad Textile Industry's Research Association (ATIRA) are pushing this frontier further by developing biodegradable polymer nano-fibres for targeted drug delivery and accelerated wound healing.

8.6 Cross-Sectoral Manufacturing Opportunity for India

8.6.1 Midstream and Downstream Value Capture

Historically, India's role in the global carbon fibre ecosystem has been restricted to the downstream segment, importing raw fibre to fabricate basic composite components. As domestic raw material becomes available, the most lucrative near-term opportunity for new entrants lies in the "midstream" segment: engineering high-margin intermediate materials such as aerospace-grade prepregs, thermoplastic tapes for electric vehicles (EVs), and specialised moulding compounds. By upgrading from manual layup processes to automated fibre placement, Indian firms like Tata Advanced Systems and Hindustan Aeronautics Limited are already integrating themselves into the global supply chains of major aerospace entities like Boeing. Expanding these midstream capabilities will empower the thousands of Micro, Small, and Medium Enterprises (MSMEs) currently dominating India's sports and industrial sectors to transition into high-value, deep-tech manufacturing.

8.6.2 Make in India Push

This manufacturing leap is being heavily catalysed by proactive government interventions under the "Make in India" and *Aatmanirbhar Bharat* (Self-Reliant India) umbrellas. To stimulate domestic production, the Government of India has sanctioned numerous R&D projects through the National Technical Textiles Mission (NTTM) and implemented robust Production-Linked Incentive (PLI) schemes across critical sectors like automobile components, drones, and Advanced Chemistry Cell (ACC) battery storage. Furthermore, recent fiscal policies, such as granting full customs duty exemptions on critical petrochemical inputs and establishing a dedicated ₹10,000 crore SME Growth Fund, are specifically designed to ease working capital burdens, lower production costs, and accelerate the scale-up of domestic MSMEs.

By successfully capitalising on this cross-sectoral demand, improving R&D capabilities, and leveraging its skilled workforce, India is perfectly positioned to transition from an import-dependent consumer to a globally competitive, export-oriented advanced manufacturing hub.

Chapter 9: Recommendations and the Road Ahead

9.1 Policy Recommendations

India's ambition to host the 2036 Olympic Games and scale its sports goods exports to \$8.1 billion necessitates a comprehensive, ecosystem-wide policy overhaul. Recent policy documents, including the NITI Aayog report on the sports equipment manufacturing sector, the *Draft National Sports Policy 2024*, and the Union Budget FY 2026-27, provide a strategic blueprint to transform the sector from a fragmented, traditional industry into a highly competitive, technology-driven global manufacturing hub. To bridge the existing technological and economic gaps, several core policy recommendations have been outlined:

9.1.1 Rationalisation of Import Duties and Customs Reforms

Policymakers should implement a permanent, long-term rationalisation of import duties on aerospace-grade carbon fibre, PAN precursor materials, high-performance resins, and EVA foams, to be maintained until domestic production capacity anchored by local facilities which reaches commercial maturity and is capable of meeting domestic demand at competitive prices. This should not be treated as a temporary exemption, as has been the case with recent petrochemical duty waivers linked to West Asian supply chain disruptions, but as a structural tariff reform with a published sunset clause tied to measurable domestic production milestones.

9.1.2 Development of Port-Proximate Greenfield Clusters

Prioritise the development of at least three port-proximate greenfield composite manufacturing clusters in coastal states, specifically Gujarat (aligned with the Hazira raw material facility), Andhra Pradesh, and Tamil Nadu, each equipped with shared infrastructure including autoclave facilities, CNC machining centres, resin infusion systems, and internationally accredited testing laboratories. These clusters should be designed as full-cycle ecosystems rather than assembly zones, enabling MSMEs to access midstream and upstream capabilities, prepreg production, fibre treatment, and mould fabrication that are currently absent from the domestic landscape. Simultaneously, the existing legacy clusters in Jalandhar and Meerut should not be abandoned but rather modernised through targeted capital subsidies for equipment upgradation.

9.1.3 Establishment of World-Class Testing and Certification Infrastructure

Establish a minimum of two world-class, internationally accredited sports equipment testing and certification centres, one in the north (proximate to the Jalandhar-Meerut cluster) and one in the south (aligned with the proposed Tamil Nadu greenfield cluster). These centres should seek accreditation from relevant international sports federations, including the International Hockey Federation (FIH), Badminton World Federation (BWF), International Tennis Federation (ITF), and World Athletics,

enabling domestic manufacturers to achieve globally recognised certification without leaving India. Certification costs for MSMEs should be partially subsidised through the Market Access Initiative (MAI) scheme with a recommended subsidy ceiling of 70% of per-SKU¹³ certification costs for enterprises with annual turnover below ₹50 crore.

9.1.4 Targeted Financial Support and the 'Brand India' Initiative

NITI Aayog's proposed allocation of ₹500 crore to a 'Brand India' sports manufacturing initiative should be operationalised with two specific mandates. First, a minimum of 30% of the fund should be directed toward co-branding and international marketing support for domestically certified composite sports equipment. Second, there should be a fair allocation within the initiative directed toward seeding domestic production of Paralympic-grade carbon fibre equipment, in partnership with the Paralympic Committee of India and manufacturers who have existing composite fabrication capabilities. Furthermore, the Draft National Sports Policy 2024's recommendation to mandate the use of 'Made in India' composite equipment at domestically hosted mega-events should be extended explicitly to include Paralympic and adaptive sports equipment, ensuring that the procurement pipeline from a potential 2036 Olympic and Paralympic Games generates guaranteed demand for domestically manufactured composite gear across all athlete categories.

9.2 Industry and Investment Actions

To fully realize India's export potential and transition the sports equipment sector towards advanced composite manufacturing, supportive government policies must be met with aggressive, strategic actions from the private industry.

9.2.1 Catalysing Domestic Raw Material Independence

Industrial conglomerates must lead backward integration into upstream carbon fibre production, treating it as a national strategic investment rather than a conventional commercial calculation. Reliance Industries' planned 20,000 MTPA carbon fibre facility in Hazira, Gujarat, and Jindal Advanced Materials committed USD 327 million investment in a 3,500 MTPA plant represent the critical first wave of this upstream capability. Industry bodies should actively facilitate technology transfer agreements and joint ventures with Tier-1 global carbon fibre producers, including Toray Industries, Teijin Limited, and Hexcel Corporation, to accelerate India's access to high-modulus fibre grades (T800 and above) that are essential for elite sports, aerospace, and defence applications and that remain subject to technology access restrictions.

¹³ SKU stands for **Stock Keeping Unit**. It is a unique, customizable alphanumeric code used by sports retailers, teams, and brands to track merchandise inventory.

9.2.2 Attracting FDI and Forging Global Joint Ventures

Domestic firms should actively leverage the current strong inflow of sports sector FDI. Nike, Puma, Adidas, and Crocs have committed a combined \$2 billion to Indian manufacturing, and Decathlon has announced a \$111 million expansion. Rather than accepting pure contract manufacturing arrangements that generate employment but no proprietary capability, Indian firms should push for co-development agreements that include composite fabrication training, quality assurance technology transfer, and joint intellectual property arrangements on India-specific product adaptations. Additionally, Taiwan's Pou Chen Group's committed ₹2,302 crore investment in Tamil Nadu expected to create 20,000 jobs should serve as a template for attracting specialist composite sports equipment manufacturers from South Korea and Taiwan, whose technical expertise in carbon fibre racket and hockey stick fabrication is directly applicable to India's priority sports manufacturing segments. Industry associations should partner with ITIs, polytechnics, and institutions like IIT Madras to develop formal certification programmes creating a pipeline of technically qualified workers that can support both FDI-led facilities and domestic MSME upgrading.

9.2.3 Traditional Craftsmanship & Deep-Tech Startups

Legacy composite manufacturers should formalise R&D partnerships with deep-tech sports startups and academic institutions, specifically targeting the integration of graphene-based e-textile sensors (being developed at ATIRA). IIT Madras has also committed to incubate 200 deep-tech sports startups over five. A portion of the proposed 'Brand India' fund should be specifically allocated to support co-development grants for manufacturer-startup-institution tripartite partnerships, with intellectual property shared equitably between commercial and academic partners. Products emerging from these collaborations particularly smart carbon fibre rackets and personalised Paralympic blades incorporating embedded performance sensors would represent a genuinely differentiated Indian offering in the global premium sports equipment market, commanding the pricing premium necessary to build brand equity beyond contract manufacturing.

9.2.4 Prioritising Sustainable Manufacturing and Circular Economy Initiatives

Indian composite manufacturers must treat sustainability compliance not as a future obligation but as an immediate action, utilising near-term working groups to establish India-specific protocols for collecting and processing end-of-life sports equipment into manufacturing feedstock via HiPerDiF-type realignment technology. This domestic push toward a circular economy is strongly reinforced by breakthrough technological precedents set at the Indian Institute of Technology (IIT) Mandi¹⁴, where researchers developed a rapid, microwave-assisted chemical recycling process using eco-friendly

¹⁴ Press Information Bureau. (2022, October 6). *IIT Mandi researchers use microwaves and eco-friendly chemicals to recover glass fibres from decommissioned wind turbine blades* [Press release]. Ministry of Information and Broadcasting, Government of India. <https://www.pib.gov.in/PressReleasePage.aspx?PRID=1865535>

hydrogen peroxide and acetic acid to achieve a 97.2% epoxy decomposition rate, recovering fibres that retain 99% of their virgin strength. Such clean-tech innovations are vital to mitigating environmental degradation, especially given that global wind turbine blades alone are projected to generate over 200,000 tons of composite waste between 2024 and 2034. Moving forward, medium-term R&D investments must transition toward bio-based precursor alternatives to polyacrylonitrile, such as lignin-based and bio-acrylonitrile carbon fibres, while government funding through the National Technical Textiles Mission must explicitly mandate sustainability and circularity as strict criteria to ensure India's emerging production capability is built on a sustainable foundation that readies manufacturers for immediate European Carbon Border Adjustment Mechanism (CBAM) compliance.

9.3 A 2030 Roadmap for Carbon Fibre in Indian Sport

To transition from a traditional, import-dependent sports manufacturing sector to a technology-driven global powerhouse, India must execute a highly structured, multi-phase roadmap. Aligning with the NITI Aayog's vision to scale sports equipment exports to \$8.1 billion by 2036 and support India's ambition to host the 2036 Summer Olympics, the following 2030 roadmap outlines the critical milestones for integrating carbon fibre and advanced composites into Indian sports manufacturing.

9.3.1 Phase 1: Achieving Raw Material Sovereignty

The prerequisite for every subsequent phase is the elimination of India's 100% dependency on imported raw carbon fibre. Without domestic raw material, no policy intervention in fabrication, certification, or branding can overcome the structural cost disadvantage that currently makes Indian composite sports equipment uncompetitive at scale.

The anchor investment for this phase is already committed: Reliance Industries' 20,000 MTPA carbon fibre facility in Hazira, Gujarat, and Jindal Advanced Materials' 3,500 MTPA plant, both scheduled to commence production starting in 2027. The National Technical Textiles Mission's 156 sanctioned research projects include indigenous carbon fibre development as a priority workstream, providing the public sector R&D complement to these private investments.

9.3.2 Phase 2: Technological Integration and 'Smart' Composites

Once affordable domestic raw material is secured, the industry must make a decisive break from the hand layup and open mould processes that currently account for 41% of Indian composite manufacturing. This phase is about closing the fabrication technology gap with East Asian competitors, not merely catching up to where China was in 2010, but building directly toward Third Generation automated and digitally integrated manufacturing.

9.3.3 Phase 3: Modernising Supply Chains and Global Branding

With domestic raw material established and Third Generation fabrication capability proven, this phase

focuses on the commercial and logistical transformation required to convert manufacturing capability into export revenue. The greenfield clusters established in Phase 2 should, by this phase, be operating at sufficient scale to demonstrate the supply chain integration advantages, reduced transit costs, faster export turnaround, and proximity to port logistics that currently give Chinese and Taiwanese manufacturers a structural advantage. Legacy clusters in Jalandhar and Meerut, having undergone modernisation investment through Phases 1 and 2, should by this phase be producing hybrid composite products that blend their traditional craftsmanship heritage with advanced carbon fibre integration a genuinely differentiated positioning that neither China nor Taiwan can replicate.

9.3.4 Phase 4: Sustainability and Circular Economy Compliance

The ultimate benchmark for this phase is India's readiness to supply domestically manufactured, internationally certified, sustainably produced composite sports and Paralympic equipment to its own athletes and to the world at a potential 2036 Olympic and Paralympic Games on Indian soil.

Chapter: 10 The Economics of Carbon Fibre: Manufacturing, Recycling, and Cost Optimisation for India's Sports Industry

While carbon fibre is globally recognised for its exceptional strength-to-weight ratio, its widespread adoption outside of the aerospace and defence sectors has historically been restricted by one major barrier: exorbitant material costs. For the Indian sports equipment manufacturing, which relies entirely on expensive imported raw materials, understanding and mitigating these costs is critical for remaining globally competitive.

To address this economic challenge, the following cost modelling analysis draws upon two foundational research studies. This includes the primary manufacturing costs and the potential of low-cost, 'heavy-tow' textile fibres¹⁵, and the economic viability of recovering carbon fibre from waste is based on the life-cycle cost (LCC) frameworks¹⁶.

10.1 Carbon Fibre Cost Modelling

The economic modelling of carbon fibre can be divided into the cost of producing virgin fibres and the cost of recovering them from waste.

10.1.1 Primary Manufacturing Cost Model (Virgin Carbon Fibre)

The traditional manufacturing of standard 50K tow polyacrylonitrile (PAN)-based carbon fibre is estimated to cost **\$17.98 per kg**. The cost breakdown is heavily skewed towards raw materials, with the PAN precursor alone accounting for 45% of the total cost (due to a low chemical yield of about 48% during conversion). Capital equipment (36.8%), labour (11.5%), and energy (6.7%) make up the rest of the expenses.

To drastically reduce these costs, researchers modelled the use of heavy-tow textile acrylic precursors (such as 457K tow) instead of expensive, specialised aerospace precursors. Because textile tows are thicker, the precursor cost drops from \$3.63/kg to \$2.24/kg, and the factory can process much higher volumes simultaneously. Under a full-utilisation scenario with optimised line speeds and reduced oven

¹⁵ SANDIA REPORT SAND2019-14173 Printed November 2019 Optimized Carbon Fibre Composites in Wind Turbine Blade Design Brandon L. Ennis, Christopher L. Kelley, Brian T. Naughton Sandia National Laboratories Robert E. Norris, Sujit Das, Dominic Lee Oak Ridge National Laboratory David A. Miller Montana State University

¹⁶ Shehab E, Meirbekov A, Amantayeva A, Tokbolat S. Cost Modelling for Recycling Fibre-Reinforced Composites: State-of-the-Art and Future Research. *Polymers* (Basel). 2022 Dec 29;15(1):150. doi: 10.3390/polym15010150. PMID: 36616500; PMCID: PMC9823720.

times, the manufacturing cost drops by 57% to just \$7.82 per kg. The relationship between manufacturing cost and market price used in this framework follows the standard composite material manufacturing margin¹⁷-

$$\text{Price} = \text{Cost} \times 1.45$$

10.1.2 Carbon Fibre Cost Sensitivity to Mechanical Properties

A further analytical output of the model¹⁸ is the derived relationship between carbon fibre manufacturing cost and its two principal mechanical properties, tensile strength (UTS in MPa) and elastic modulus (E in GPa) *¹⁹:

$$\text{CF Cost (\$/kg)} = -6.45 + 0.005 \times \text{UTS} + 0.016 \times E$$

This linear regression relationship ($R^2 > 0.99$, $n = 40$ observations) reveals that tensile strength is a significant cost driver: higher precursor quality required to achieve greater tensile strength carries a substantial cost premium.

10.1.3 End-of-Life Cost Model (Recycling)

Because virgin carbon fibre is energy-intensive and costly to produce, Life-Cycle Cost (LCC) models are used to evaluate the economic feasibility of recycling compared to traditional landfilling. The viability is measured by the Unit Cost per mass of recovered Fibre (UCF):

- *Mechanical Recycling (\$0 - \$3.5/kg UCF)*: The cheapest method, but it severely degrades the fibre into short pieces or powder, meaning it cannot be used for high-performance structural applications.
- *Pyrolysis (\$1.6 - \$5.5/kg UCF)*: Uses heat to burn away the resin. It retains fibre length, but the tensile strength drops unpredictably, retaining anywhere from 4% to 85% of its virgin strength.
- *Solvolyis / Chemical Recycling (\$14 - \$28/kg UCF)*: The most expensive method, but it yields the highest quality material, retaining over 90% of the virgin fibre's strength.

¹⁷ This multiplier, derived from EPA automobile industry cost studies and applied by Ennis et al. (2019), accounts for selling, general, and administrative expenses and profit margin, and is used throughout this analysis to convert estimated manufacturing costs into indicative market prices.

¹⁸ Ennis, B. L., Kelley, C. L., Naughton, B. T., Norris, R. E., Das, S., Lee, D., & Miller, D. A. (2019). *Optimized carbon fibre composites in wind turbine blade design* (Report No. SAND2019-14173). Sandia National Laboratories.

¹⁹ UTS stands for Ultimate Tensile Strength.

Measured in MPa (Megapascals)

It represents the maximum amount of pulling or stretching force a material can handle before it cracks, breaks, or pulls apart completely.

10.2 Strategic Adaptation for the Sports Manufacturing Sector in India

To apply these cost models to the Indian sports equipment manufacturing sector, several domestic parameters and economic challenges must be factored in.

10.2.1 Overcoming Import Dependency and Cost Disadvantages

Currently, India does not produce any raw carbon fibre domestically and relies entirely on imports from countries like Japan, the US, and France. Because of this, Indian sports equipment manufacturers face a cost disadvantage compared to manufacturing hubs like China, primarily due to high import duties levied on critical raw materials like carbon fibre, polyurethane, and EVA foam.

10.2.2 Applying the Heavy-Tow Cost Model to Indian Sports Goods

To remain competitive and achieve the NITI Aayog's ambitious target of \$8.1 billion in sports equipment exports by 2036, Indian manufacturers must strategically adapt the heavy-tow cost model. The cost-to-property formula ($-6.45 + 0.005 \cdot \text{UTS} + 0.016 \cdot \text{E}$) is highly relevant here: sports equipment (like hockey sticks, badminton racquets, and bicycle frames) requires excellent stiffness and vibration dampening, but rarely requires the extreme tensile strength needed for ballistic missiles or fighter jets. By purposely pivoting to import lower-cost, heavy-tow textile carbon fibres, which sacrifice excess tensile strength but maintain stiffness, Indian MSMEs can cut their raw material expenses by nearly half.

10.2.3 Leveraging Recycled Carbon Fibre for Mass-Market Sporting Goods

While global recycling models show that recovering carbon fibre struggles to compete with cheap landfilling, the Indian perspective is different due to the inflated cost of imported virgin materials. Indian manufacturers can utilise localised Pyrolysis (costing -\$1.6 to \$5.5/kg) to recover carbon fibres from aerospace or wind turbine waste. Even though these fibres lose some tensile strength, they are highly suitable as reinforcing fillers for mass-market sporting goods (such as protective helmet shells, shin guards, and entry-level racquets).

10.2.4 Future Domestic Outlook

This reliance on imported cost-modelling strategies will serve as a vital bridge until India's upstream manufacturing matures. With companies like Reliance Industries building a massive 20,000 Metric Tonnes Per Annum (MTPA) carbon fibre facility in Gujarat, and the government pushing for domestic

production by FY 2025–26, India is gradually moving towards establishing its own localised, cost-effective carbon fibre supply chain.

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Annexure

Kobayashi et al. (2026) conducted a meta-analysis of 14 trials confirming that carbon-plated running shoes reduce metabolic demand by a mean of 2.88%, translating to an estimated 1% improvement in overall marathon performance a margin that, at the highest level of competition, is decisive. From consumer performance footwear to dynamic prosthetic blades for Paralympic sprinters, advanced composite materials have moved beyond passive equipment design to become active contributors to athletic performance.

Appendix Table 1- *Physiological Efficiency Outcomes (Mean Differences)*

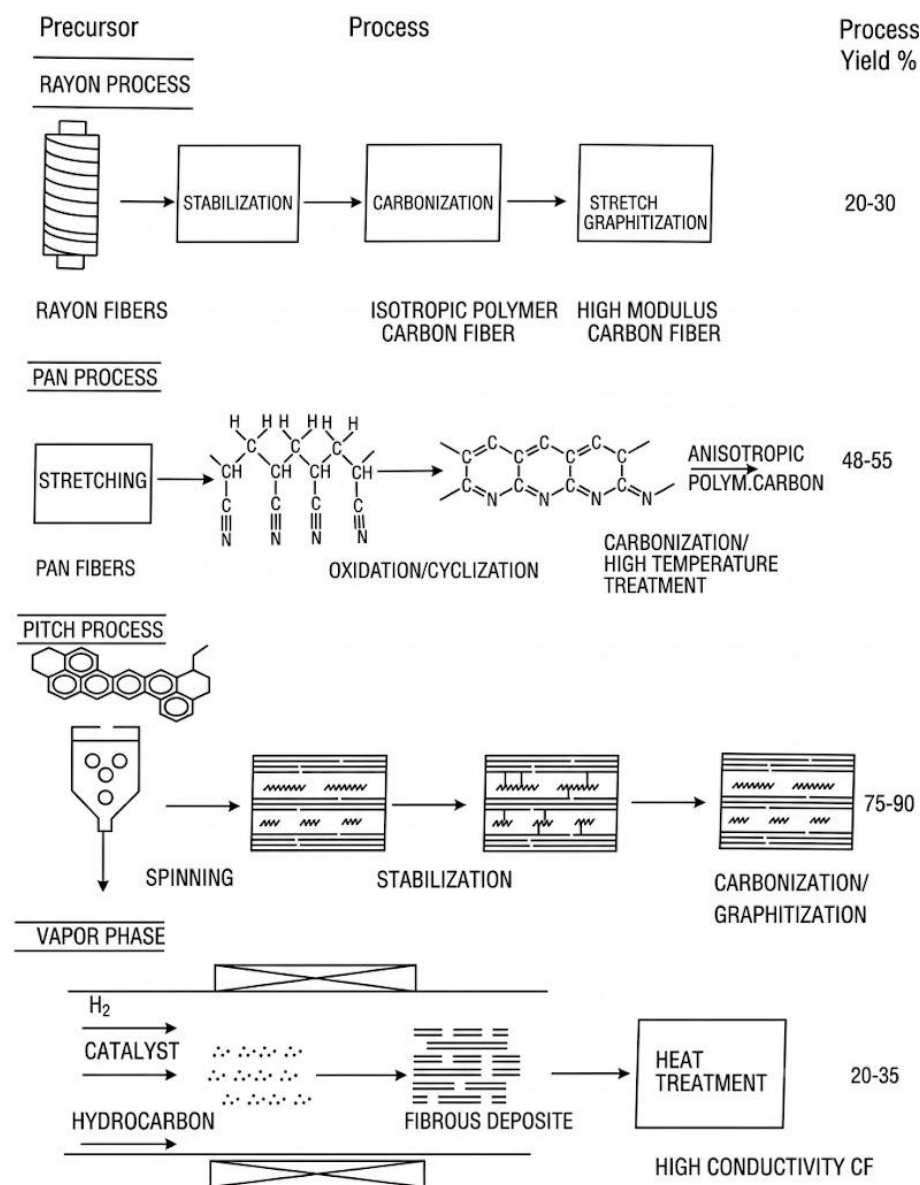
Parameter	Mean Difference %
Running Economy	−2.88%
Metabolic Cost	−2.64%
Oxygen Consumption	−2.84%
Energetic Cost of Transport	−2.62%
Mean Across-Outcome	−2.75%

Negative values indicate lower metabolic demand (improved efficiency) when using the carbon-plate condition.

Source: Adapted from Kobayashi, E. N., de Toledo, R. R. F., de Almeida, M. O., Sprey, J. W. C., & Jorge, P. B. (2026). Metabolic effects of carbon-

Annexure

Figure 24 Schematic comparison of processing routes for carbon fibres

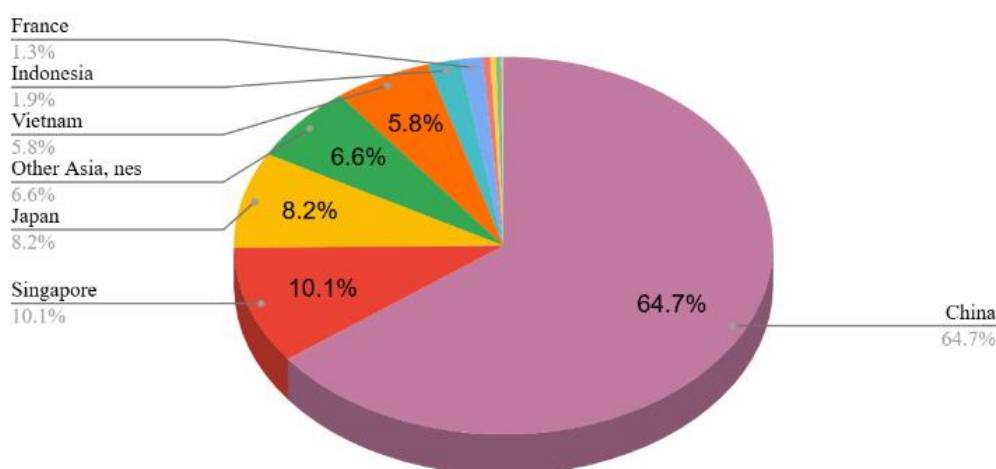


Source: <https://www.sciencedirect.com/topics/engineering/carbon-fibre-production>

Annexure

1. India's imports of Badminton or similar rackets, whether or not it was \$26,902.94K and quantity 110,080,000 items. India imported Badminton or similar rackets, whether or not st from China (\$17,415.36K, 109,455,000 Item), Singapore (\$2,709.08K, 208,431 Item), Japan (\$2,217.08K, 44,055 Item), Other Asia, nes (\$1,767.61K, 142,274 Item), Vietnam (\$1,570.62K, 136,110 Item)

India Badminton or similar rackets, whether or not st imports by country in 2024 (HS Code: 950659)



Source: WITS -HS Code 950659: Badminton or similar rackets, whether or not st

2. India imports of Lawn-tennis rackets, whether or not strung was \$4,503.70K and quantity 293,112 Item. (HS Code- 950659)
3. India imports of Bicycle frames and forks, and parts thereof, was \$21,268.33K. (HS Code- 871491)
4. India imports of Art.&equip for sports&outdoor games nes&swimmin was \$45,747.27K and quantity 44,271,700 Item. (HS Code 950699)

Annexure

Toray Carbon Fibre Performance Comparison

Carbon Grade	Tensile Strength (MPa)	Tensile Modulus (GPa)	Primary Characteristic
Toray T700	4,900	230	High Impact Resilience
Toray T800	5,490	294	Balanced Performance
Toray T1000	6,370	294	Maximum Stiffness-to-Weight